Fine-Tuning versus Retrieval-Augmented Generation in Large Language Models: A Comparative Study on Dungeons & Dragons

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***Abstract*—Domain-specific question answering often hinges on whether to retrieve external knowledge or retrain model parameters. Prior work contrasts Retrieval-Augmented Generation (RAG) with Fine-Tuning (FT) in generic tasks, but creativity-laden, complex settings like Dungeons & Dragons (D&D) remain under-explored. We conduct a comparative study of four configurations (Base, FT, RAG, and FT+RAG) on consumer-grade hardware, evaluating 1120 generated answers against a 280-item test set using a dual-blind expert grading rubric and a suite of nine automated Ragas metrics.**

Our findings show conclusively that RAG is the single most impactful strategy for factual accuracy. The RAG model achieved an 81% improvement in expert-judged correctness over the baseline, while standalone FT provided no significant gains. The hybrid FT+RAG model yielded the highest semantic similarity to ground-truth answers, demonstrating a synergistic effect. A key takeaway is that a powerful base LLM can successfully reason over a broad, high-recall context, providing accurate answers even when specific automated metrics like Context Relevance and Response Groundedness score low due to evaluation artifacts. This work validates the feasibility of creating a reliable, high-performing D&D assistant on consumer hardware and underscores the importance of pairing high-recall RAG with a strong reasoning model for complex domains. The FT model can be found on: https://huggingface.co/astrevallion/Qwen3-14B-FT and the GitHub repository with all the source code, full appendices and documents can be found on: https://github.com/ErcagCan/ft_vs_rag_dnd.

1. Introduction

LLMs are poised to become indispensable assistants at the gaming table, yet today's models still struggle with the two abilities Dungeon Masters (DMs) value most: (i) factual fidelity regarding the D&D ruleset and (ii) the imagination required to weave engaging stories with those rules. The SRD 5.1 provides an open 403-page legal subset of the core rules [1]. A reliable, creative set of rules applied in a complex context, which can be accessed fairly quickly, could therefore benefit millions of hobbyists, content creators and VTT platforms.

1.1. Motivation

1.1.1. From rulebooks to cloud services. In a wide-ranging interview with The Verge, Hasbro CEO Chris Cocks describes how D&D has evolved from a paperback rule-set into a "play-everywhere" trans-media property that now spans films, streaming shows, virtual-table-top (VTT) clients and the official D&D Beyond rules compendium [2]. Cocks notes that almost every major business decision at Wizards of the Coast, Hasbro's games division, now passes through a digital lens:

- the 2022 acquisition of D&D Beyond for $147 million formalised an always-online rules portal and character builder
- a forthcoming Unreal-based 3-D VTT is being tested internally to make remote play "feel like a miniature war-game on a living table"
- a $100 million "brand-insights" platform is being built to mine play telemetry and social chatter for product decisions.

Taken together, these moves signal a structural shift: rule adjudication and narrative support are no longer bound to a physical table or a single DM but are increasingly delegated to cloud-native services.

1.1.2. A ten-million-player problem. The scale of that shift is already visible in the broader ecosystem. The largest VTT, Roll20, doubled its user base from 5 million to 10 million between March 2020 and February 2022 [3]. More than half of the sessions hosted on the platform run D&D 5E, and each session places continuous, real-time demands on rules lookup, encounter balancing and improvisational storytelling. At this order of magnitude a single mis-interpreted rule propagates quickly through social media, actual-play streams and home games. Reliable, scalable, and creatively flexible rules assistants are therefore becoming a critical usability layer for both official and third-party VTTs.

1.1.3. Why language-model adaptation matters. LLMs already demonstrate near-human fluency, but tabletop gaming stresses two capabilities that remain brittle:

- Factual fidelity to an intricate rules corpus
- Narrative inventiveness and the ability to weave those rules into engaging, context-aware story hooks, NPC dialogue and encounter descriptions.

Two broad adaptation strategies dominate the literature:

- PEFT (e.g., LoRA/QLoRA) internalizes domain knowledge and style but risks overfitting and stale facts.
- RAG looks up facts in an external database before prompting. Augmenting the found facts to the prompt keeps the model from hallucinating, but this can make the generated answer feel correct yet narratively flat (e.g., stat-block regurgitation).

Recent empirical work underscores the potential and limits of LLMS for tabletop role-playing. In their CALYPSO study, Zhu et al. (2023) found that off-the-shelf GPT-3.5-turbo can already take on several DM duties, generating concise monster descriptions, brainstorming encounters, and producing ready-to-present prose that most DMs found usable. However, the summarise-then-generate pipeline still produces frequent hallucinations: the model invents monster abilities, misstates rules, or otherwise departs from the source text. While the authors note occasional tonal inconsistencies, they do not quantify stereotyping or tonal drift. The authors explicitly call for domain-specific, parameter-efficient fine-tuning (and other lightweight adaptation methods) to curb these errors and preserve narrative voice. [4]

In a pilot investigation of ChatGPT as a sole DM, Triyason (2023) reported that the vanilla model achieved high ratings on narrative coherence (4.3/5) and adaptability (4.3/5) but a lower score on immersion (2.7/5). The authors attributed the lower immersion to the model's reliance on generic prose and the absence of game-specific lore, classic symptoms of insufficient in-domain grounding. Players described the experience as ”disappointing” once these shortcomings became apparent. Triyason recommends deploying domain-specific adapters (e.g., fine-tuned prompts or embeddings that encode relevant lore) before broader deployment, particularly to aid newcomers who may lack rules literacy. [5]

Taken together, these findings sharpen the motivation for our head-to-head comparison. They demonstrate that unadapted LLMs already deliver partial value, but that factual fidelity and stylistic authenticity remain brittle in rule-dense, creativity-laden domains like D&D. Systematically contrasting FT, RAG and their hybrid therefore addresses not only an engineering question (latency versus quality) but an emerging responsibility gap in AI-augmented tabletop play.

However, no published study has compared these strategies head-to-head under the hardware and latency constraints faced by consumer-grade VTT integrations. As the D&D player base crests eight figures and official content pipelines accelerate, understanding which adaptation path yields the best balance between rule accuracy and creative value is no longer an academic curiosity. This has become an operational prerequisite for the next generation of digital play tools.

> This work therefore asks (research question):
> Out of FT, RAG, or their combination, which adaptation strategy delivers the most reliable and inspiring D&D assistant when both GPU memory and response time are strictly limited?

Answering this question will inform not only AI-augmented refereeing for VTTs, but also broader mixed-initiative design tools for story-driven games and educational simulations.

1.1.4. Research question contributions. To find an answer to the research question defined in the motivation section, three general steps have been taken:

- **Construct four configurations**: Base, +RAG, +FT, and +FT +RAG.
- **Curate** a balanced D&D query set spanning rules adjudication, narrative hooks, and monster design, each rooted in the SRD.
- **Analyse** quantitative gains highlighting potentially how RAG curbs rule hallucinations while FT polishes tone and how their synergy delivers the best of both worlds.

1.1.5. Solution Requirements. The comparative study is guided by two non-negotiable **system requirements**, derived from the realities of live virtual-table-top (VTT) play and the quality expectations of DMs:

Latency Requirements: Our latency requirements are not arbitrary but are grounded in foundational Human-Computer Interaction (HCI) research that establishes critical thresholds for human cognitive processing. For this study, we define **interactive latency** as the time from query submission to the appearance of the first token in the response stream. This initial delay is the period during which the user is actively waiting, and it is this duration that the cognitive thresholds identified by Miller (1968) directly address.

Miller's work distinguishes between different psychological boundaries for different interaction types:

1) **The Conversational Threshold ($<$ 2 seconds):** For seamless, back-and-forth interactions, Miller notes that delays exceeding two seconds disrupt the user's immediate ”thread of communication” [6]. We apply this strict criterion to our **Base** and **FT** models. These configurations are expected to handle direct queries in a fluid, conversational manner, akin to a quick clarification. Therefore, their time to first token must be **less than 2 seconds** to maintain this rapid interaction style.

2) **The Problem-Solving Threshold (2-10 seconds):** For more complex tasks, Miller establishes a critical breakpoint for user attention, stating that "response delays of a standard ten seconds will not permit the kind of thinking continuity essential to sustained problem solving" [6]. This rationale perfectly applies to our **RAG** and **FT+RAG** configurations. These models perform an explicit RAG step to answer more nuanced queries, and a DM using them is engaged in this kind of problem-solving. We therefore stipulate that their time to first token must fall within the **2 to 10-second** range, acknowledging the added complexity while strictly adhering to the maximum cognitive limit required to maintain focus and game immersion.

Any configuration failing its respective threshold is thus considered unsuitable for its intended interaction style at a live gaming table.

Table-Ready Reliability: For a VTT assistant to be a help rather than a hindrance, it must possess what we term "table-ready reliability." This goes beyond simple correctness; the model's output must be both factually sound and demonstrably grounded in evidence to a DM managing a live game. The literature consistently identifies factual errors and hallucinations as primary points of failure that erode user trust and disrupt gameplay [4], [5]. Therefore, we establish a composite requirement with two core principles:

1) **Significant Improvement in Factual Accuracy:** An adapted model configuration must provide a statistically significant improvement in factual correctness over the unassisted baseline model. An assistant that is not demonstrably more accurate than the base LLM offers no practical value for rule adjudication. This principle is evaluated through the "Correctness" dimension of our expert-led Likert scoring, where we look for a clear and substantial increase in the mean score for any adapted configuration (FT, RAG, or FT+RAG) when compared to the BASE model.
2) **Evidenced-Based Grounding:** For any configuration employing RAG (RAG and FT+RAG), the generated answer must be supported by the retrieved context. This principle ensures that the model is "showing its work," allowing a DM to verify the source of a ruling and build confidence in the system. We quantify this using the **Response Groundedness** metric from the Ragas framework, which directly measures the proportion of claims in the answer that can be verified against the provided context. A successful RAG system must demonstrate a tangible, measurable score in this dimension.

A configuration must satisfy both of these conditions to be considered a viable VTT co-DM. Failure to significantly improve accuracy or to ground its responses in evidence renders the tool unreliable for use in a live play environment.

We treat the requirements as *hard constraints*. A configuration that fails either bound, regardless of stylistic flair, does not qualify as a viable VTT co-DM.

2. Related Work

Recent literature converges on two complementary strategies for adapting LLMs to specialised domains: PEFT (e.g. LoRA/QLoRA with Chain-of-Thought (CoT) traces) and RAG. "Thinking-type" models such as Qwen3-14B and gpt-oss-20b offer a suitable small-scale backbone by exposing an explicit reasoning mode without the infrastructure costs of 70 B-parameter or larger models. Meanwhile, lightweight RAG pipelines (LightRAG, GraphRAG) and agentic variants (ReAct, Retriever Agents) increasingly promise hallucination-free answers. Recent work compares FT and RAG on existing public datasets. However, no published study has compared fine-tuned, RAG-enhanced, and hybrid adaptations on the small thinking models, creative data and expert-centred creativity rubric like this project strives to achieve.

2.1. Thinking-type LLMs

Early open-source LLMs (LLaMA-2, Phi-2) targeted raw perplexity, but Qwen3 and gpt-oss:20b are accompanied by a thinking switch that yields multi-step reasoning gains. Similar dual-mode control appears in more and more recent LLMs, yet no study benchmarks these small-size models on creative tabletop-gaming tasks. Their moderate 14-34B parameter footprint makes them ideal candidates for adaptation on consumer-grade hardware. This feasibility is directly supported by the work of Dettmers et al. on QLoRA, who demonstrated that it is possible to fine-tune a 33B parameter model on a single consumer GPU with 24GB of VRAM, a hardware profile matching the RTX 4090 used in this study [7].

2.2. Parameter-efficient FT

Parameter-efficient FT (PEFT) methods adapt large models without retraining all weights. LoRA (Low-Rank Adaptation), for instance, freezes the pre-trained model and injects small, trainable rank-decomposition matrices, reducing the number of trainable parameters by as much as 10,000 times and GPU memory requirements by up to 3 times [8]. QLoRA builds on this by quantizing the frozen backbone model to 4-bits, which enables the FT of massive 65-billion parameter models on a single 48GB GPU in under 24 hours [7].

Furthermore, the reasoning capabilities of smaller models can be significantly enhanced by fine-tuning them on CoT outputs generated by a larger "teacher" model. This knowledge distillation has been shown to enable a fine-tuned 7B parameter model to rival the performance of GPT-3 175B on mathematical benchmarks [9]. However, this prior work has largely focused on uniform reasoning datasets like GSM8K, leaving a gap in understanding its effectiveness on lore-heavy, semi-creative corpora such as the SRD.

2.3. The Canonical RAG model in Detail

RAG is a hybrid framework that equips LLMs with a non-parametric memory, allowing them to consult an external knowledge base at inference time while still leveraging the rich parametric knowledge stored in their weights. In contrast to purely generative models, whose knowledge is frozen at training time, RAG retrieves task-relevant passages on-the-fly and conditions the generator on this evidence, yielding answers that are both more factual and easier to update [10].

2.3.1. Core Architecture. The original RAG model couples two pre-trained components:

- **Retriever (p_η):** A dense bi-encoder, such as Dense Passage Retrieval (DPR), that embeds the input query and uses Maximum-Inner-Product-Search (MIPS) to find the top-K most relevant documents from a vector index (e.g., Wikipedia).
- **Generator (p_θ):** A sequence-to-sequence model, such as BART, which takes the original input and the retrieved passages and autoregressively generates the final output sequence.

By treating the retrieved document as a latent variable, the entire model can be fine-tuned end-to-end, allowing both the retriever and generator to align on the downstream task [10].

2.3.2. Empirical Benefits. On open-domain QA benchmarks, the RAG framework was shown to match or surpass the performance of state-of-the-art parametric-only models while using far fewer parameters. The key advantages demonstrated were:

- **Faithfulness:** Generated answers contain fewer hallucinations and more specific, factual details than baselines.
- **Interpretability:** The retrieved passages serve as explicit evidence for the model's claims, providing a degree of transparency.
- **Updatability:** The model's world knowledge can be instantly refreshed by simply swapping the document index, without any need for expensive retraining or gradient updates [10].

2.3.3. Advances with Off-the-Shelf Retrievers. While the original RAG formulation jointly fine-tuned the retriever for each downstream task, subsequent work has shown this is not strictly necessary for strong performance. Neelakantan et al. (2022) demonstrated that replacing the retriever with a large, off-the-shelf encoder pre-trained with a contrastive objective significantly boosts RAG quality. On the Natural Questions benchmark, this approach raised Recall@20 from 67.2 to 78.8 without any task-specific training, thereby simplifying deployment and reducing computational overhead [11]. This highlights a shift towards using powerful, general-purpose encoders as the RAG backbone in RAG-like systems.

2.3.4. Key Training Considerations. The work by Neelakantan et al. (2022) also highlighted a key factor for achieving high performance with contrastive pre-training: **batch size**. Because the method relies on in-batch negatives, they demonstrate that using a sufficiently large batch (e.g., over 12,000 examples) is crucial for providing enough hard negatives to the model, which results in a significant performance boost.

2.4. RAG Variants

The term RAG spans a diverse toolkit that ranges from simple BM25 look-ups to full agentic planners. We therefore organise prior work into five technical strata, *dual-encoder baselines, fusion models, memory-augmented architectures, graph-enhanced retrievers,* and *agentic RAG loops* and summarise the mechanics of each.

2.4.1. Dual-encoder Baselines. The initial wave of RAG models was defined by two pioneering dual-encoder architectures that established different training paradigms.

The canonical 'RAG' model, introduced by Lewis et al. (2020), pairs a pre-trained DPR retriever with a pre-trained BART-Large generator. In this framework, the model is fine-tuned end-to-end on a downstream task. For a given input, the retriever finds the top-K relevant passages, and the generator uses these passages as context to produce the final output. The key innovation was a FT recipe that allowed the model to marginalize over the retrieved documents as latent variables, achieving strong performance on open-domain QA [10].

In parallel, 'REALM' (Retrieval-Augmented Language Model) by Guu et al. (2020) focused on retrieval-augmented *pre-training*. REALM was the first to demonstrate that a retriever could be trained from scratch in an unsupervised manner. It jointly pre-trains the retriever and the language model, using the masked language modeling objective as the learning signal. This allows the token-likelihood gradients to flow back through the RAG step, effectively teaching the retriever which documents are useful and directly "sculpting" the knowledge index space [12].

2.4.2. Fusion Models. A key limitation of early RAG models was the computational bottleneck created by concatenating all retrieved passages before feeding them to the encoder. To address this, Izacard & Grave (2020) introduced Fusion-in-Decoder (FiD), an architecture that processes each passage-query pair independently in the encoder. A "late fusion" is then performed, where the decoder cross-attends to the aggregated representations of all passages. This more efficient approach proved highly effective, lifting the Exact Match score on Natural Questions to 48.2, a significant gain over the 44.5 achieved by the original RAG model [13].

Building directly on the FiD architecture, ATLAS (Izacard et al., 2022) introduced a crucial pre-training phase that jointly trains the retriever and the language model. By using objectives like Perplexity Distillation, ATLAS allows a learning signal to back-propagate from the language model to the retriever. This effectively teaches the retriever to ”choose its own evidence” by prioritizing documents that are most useful for the language model’s final prediction, leading to state-of-the-art few-shot performance [14].

2.4.3. Memory-augmented transformers. A key architecture in RAG systems is DeepMind’s **RETRO** (Retrieval-Enhanced Transformer). It enhances a standard auto-regressive language model by pairing a parametric ”reasoner” model (up to 7.5 billion parameters) with a massive, non-parametric external memory. This external key-value store is built from a corpus of over 2 trillion tokens, requiring approximately 93 terabytes of disk space to index.

At generation time, RETRO splits the input sequence into 64-token chunks. For each chunk, it retrieves the nearest ”neighbours” from the database and uses a specialized **chunked cross-attention mechanism** to incorporate this external knowledge into its predictions. This hybrid approach proved highly efficient, achieving performance on par with models like GPT-3 while using **25 times fewer parameters** and significantly less pre-training compute. The success of RETRO demonstrates that augmenting a model with an explicit, scalable memory can be a more efficient path to high performance than raw parameter scaling alone.

2.4.4. Graph-enhanced RAG. Classic vector search can struggle to capture multi-hop evidence spread across multiple documents. To address this, recent work augments RAG with an explicit knowledge graph, a technique broadly known as GraphRAG [15]. In this approach, documents are decomposed into entity nodes and relation edges, creating a structured index that preserves complex, inter-document relationships [15]. At inference time, graph-guided RAG algorithms traverse this index to surface interconnected facts that collectively answer a query. This method has proven effective for compositional question answering, where synthesizing information from multiple sources is critical [15].

The practical benefits of this approach are demonstrated in implementations like NVIDIA’s GraphRAG, which leverages graph-based techniques to enhance question-answering accuracy by constructing knowledge graphs from text and identifying key relationships that might otherwise be missed. NVIDIA Blogpost

However, many industrial GraphRAG systems can be computationally intensive, particularly when handling updates or large-scale queries, as they often rely on community detection and summary generation that can be costly to maintain [16]. This has inspired the development of more lightweight variants, such as ’LightRAG’, which aim to provide similar benefits with lower latency and resource requirements. LightRAG employs a dual-level RAG system over a graph index, optimizing for both efficiency and rapid adaptation to new data without needing to rebuild the entire graph structure [16].

A list of foundational GraphRAG and knowledge-graph papers is maintained at https://graphrag.com/appendices/research/.

2.4.5. Agentic RAG loops. Early tool-augmented systems like WEBGPT demonstrated that Large Language Models could be fine-tuned to interact with external environments, such as a web browser, to answer complex questions [17]. These models learn to execute a sequence of actions (e.g., SEARCH, CLICK, QUOTE) based on observations from the environment to gather evidence for an answer.

Later work, most notably the REACT framework, made the model’s reasoning process explicit by introducing an *agentic* pattern that synergizes reasoning and acting [18]. In this paradigm, the LLM generates an interleaved stream of **Thought** (a reasoning trace), **Action** (a command for an external tool), and **Observation** (the result from the tool). The model repeats this cycle until it has enough information to issue a final answer. Unlike single-shot RAG, which retrieves information once, these iterative loops can:

- **Refine queries on the fly**, allowing later searches to build on evidence found in earlier steps. This dynamic strategy improves performance on multi-hop datasets like HOTPOTQA where answers must be synthesized from multiple sources [18].
- **Recover from failed actions**, such as an empty search result. By reasoning about the failure, the model can reformulate its query and try an alternative approach [18].
- **Trade latency for accuracy** by adjusting a step budget. The number of thought-action cycles can be limited to control for response time, or expanded for more thorough investigation.
- **Expose a transparent reasoning trace**. The explicit thoughts provide a human-interpretable record of the model’s strategy, which is invaluable for debugging, auditing, and building trust in the system [18].

Together, these ”Agentic RAG” loops blur the line between classical RAG and tool-driven autonomy, allowing the model to dynamically *choose and verify its own evidence* rather than relying on a single, static RAG step.

3. Methodology

3.1. Data Acquisition and Preparation

3.1.1. Data for RAG and Synthetic Test Set (STS) generation. The initial phase of this project involved sourcing a clean, machine-readable version of the D&D 5.1 SRD. Our first attempts centered on extracting text directly from the official SRD PDF document. We experimented with several methods, including Optical Character Recognition (OCR) and various Python libraries such as `PyMuPDF`. However, these approaches consistently yielded subpar results. The extracted text was often corrupted with formatting artifacts, missing characters, and extraneous symbols, likely due to the PDF’s custom fonts and complex two-column layout. Furthermore, elements spanning the full page width, like tables and section headers, were frequently misplaced within the text, disrupting the document’s logical flow and rendering it unsuitable for downstream NLP tasks.

To overcome these challenges, we pivoted to a pre-formatted plain text version of the SRD, available under a Creative Commons license [19]. This file provided the complete and correctly structured text, serving as the single source of truth for all subsequent stages. This document was ingested by LightRAG to construct the knowledge graph entities and relations for the RAG system.

For the generation of our STS, this comprehensive text file required preprocessing. To create thematically coherent question-answer pairs using the Ragas framework, we manually segmented the entire document into smaller, topically-focused excerpts. Each excerpt was curated to be approximately 25,000 characters in length, a size conducive to generating a focused set of 4-6 high-quality questions, as detailed further in the Ragas evaluation section.

3.1.2. Data Preparation for FT. The dataset for FT the base model was prepared in a two-stage process. First, we used an open-license JSON dump of the SRD [20]. A custom script, `srd_preprocessing.py`, traversed this nested JSON structure, flattening the content, normalizing tables, and removing legal boilerplate and appendix noise. This process yielded 1,779 distinct chat pairs, where the "user" content corresponds to a breadcrumb path (e.g., "Spells > Fireball") and the "assistant" content is the relevant rule text.

Second, this initial chat file was enhanced through a CoT augmentation step using the `augment.py` script. For each of the 1,779 answer pairs, the Base LLM model was prompted to generate two additional components: (1) a synthetic, more natural-sounding user question based on the answer and its keywords, and (2) a reasoning trace enclosed in <think> tags that emulates the model's process for arriving at the answer. This augmentation aligns the training data with the desired output format, encouraging the model to expose its reasoning process before providing a final answer.

3.2. FT Pipeline with Unsloth

For our FT pipeline, we chose the Unsloth library, a toolkit designed to significantly accelerate and reduce the memory footprint of training LLMs. This choice was critical for the feasibility of our study, as it allowed us to perform full FT runs on a single consumer-grade NVIDIA RTX 4090 GPU with 24GB of VRAM. Unsloth achieves this through highly optimized kernels and memory management techniques that can reduce VRAM usage by up to 80% without sacrificing accuracy. This efficiency makes FT accessible not only on high-end local hardware but also on free Google Colab instances, democratizing the process for a wider range of researchers and developers. It is worth noting, however, that setting up the environment on Windows can present significant challenges, as the library and its dependencies are primarily optimized for Linux-based systems.

Our methodology was adapted from the official Unsloth notebook for the *qwen3:14b* model [21]. The process begins by loading the base model using Unsloth's 'FastLanguageModel', with a crucial parameter 'load_in_4bit = True'. This enables 4-bit quantization via QLoRA, a technique that dramatically reduces the model's memory footprint by representing its weights with only 4 bits, allowing the 14-billion-parameter model to fit comfortably within our VRAM limits.

Following this, we applied Low-Rank Adaptation (LoRA) using the 'get_peft_model' function. Instead of training all 14 billion parameters of the model, LoRA freezes the pre-trained weights and injects small, trainable "adapter" matrices into specific layers of the transformer architecture (e.g., 'q_proj', 'k_proj', 'v_proj'). This PEFT approach means we only need to train a tiny fraction of the total parameters (in our case, 0.86%), which drastically reduces computational cost. Key hyperparameters for this process included:

- **r = 32:** The rank, or dimensionality, of the LoRA adapter matrices. This determines the number of trainable parameters.
- **lora_alpha = 32:** The scaling factor for the LoRA weights, which we set equal to the rank as is common practice.

The training itself was orchestrated by the 'SFTTrainer' (Supervised FT Trainer) from the TRL library. We conducted numerous experiments, modifying various hyperparameters such as learning rate, rank, and batch size. However, we consistently found that deviations from the Unsloth notebook's well-tuned defaults resulted in a less stable and less effective training process, often leading to a significantly worse training loss curve. Consequently, we adhered to the optimized configuration from the notebook, which produced the stable loss convergence shown in Table 7.

3.3. Ragas: A Framework for Evaluating RAG Systems

To systematically evaluate the performance of our RAG configurations, we employ Ragas, a framework capable of assessing the quality of RAG pipelines without relying on ground-truth human annotations [22]. Ragas provides a suite of automated metrics that measure distinct aspects of the generation process, from the relevance of retrieved context to the factual accuracy and overall quality of the final generated answer. By using this framework, we can obtain a multi-faceted and objective measure of how well our different models perform in key areas critical to building a reliable D&D assistant.

3.3.1. STS Generation. To create a robust evaluation dataset without extensive manual annotation, we utilized the STS generation capabilities of Ragas in `ragas_sts_generation.py`. This code is inspired by the brief example from Ragas itself found on this link: Ragas STS Guide. This process begins by ingesting our source documents, in this case, text excerpts from the SRD. For each document, a `KnowledgeGraph` is constructed to represent the information structurally. This graph is then enriched by applying a series of transforms, including a `HeadlinesExtractor` and a `KeyphrasesExtractor`, which identify and isolate salient pieces of information from the raw text.

With the structured knowledge in place, `QuerySynthesizers` generate questions targeted at these extracted headlines and keyphrases. To ensure the generated questions reflect realistic user queries, we defined three distinct `Personas`. The final output is a STS json file containing 4-6 question-context-expected answer pairs, which serves as the foundation for our automated evaluation and the questions we will query the different configurations of our LLM system. It takes approximately 1-2 minutes to generate one file.

3.3.2. Persona-Driven Question Generation. To ensure our STS reflects the diverse needs of the D&D community, we employed a persona-driven approach to question generation. Instead of creating generic queries, we defined three distinct user archetypes, each with a specific motivation and level of expertise. This method guides the language model to generate questions that are not only topically relevant but also stylistically and intentionally aligned with realistic user scenarios. The three personas prompts are defined in the Appendix, and are described as follows:

- **The Novice:** This persona represents a DM who is familiar with the basics but still requires guidance. Their generated questions are designed to elicit step-by-step explanations of mechanics, simple clarifications of rarely used rules, and foundational lore details. They test the model's ability to provide clear, educational, and comprehensive answers.
- **The Veteran:** This persona embodies an experienced DM who values speed and precision. The questions generated from this archetype are concise and targeted, seeking authoritative rule citations, quick fact-checks, and nuanced lore details without introductory fluff. This persona tests the model's accuracy and its ability to retrieve specific, factual information efficiently.
- **The Narrativeist:** This persona focuses on the storytelling aspect of D&D. Their queries are less about mechanical adjudication and more about creative application. They ask for contextual background, flavourful details, and ways to weave rules into the narrative to enhance the story. This persona evaluates the model's capacity for creative and context-aware generation.

Together, these three personas cover the primary modes of interaction with the SRD: learning the rules (Novice), arbitrating them (Veteran), and adapting them for storytelling (Narrativeist). This ensures our evaluation is grounded in a representative sample of real-world use cases.

3.3.3. Query Synthesizers and Complexity. To comprehensively evaluate the RAG system's ability to handle varying levels of complexity, we leveraged different query synthesizers provided by the Ragas framework. These synthesizers generate questions that require different reasoning and RAG strategies, primarily distinguished as either "single-hop" or "multi-hop". The generation process is underpinned by a two-stage mechanism: knowledge graph construction and LLM-based synthesis.

First, as illustrated in Figure 1, the source documents are transformed into a structured knowledge graph. Key information such as entities and topics are extracted from the text to form nodes. A relationship builder then establishes connections between these nodes based on semantic similarity and entity overlap, creating a rich, interconnected representation of the document's content.

Second, this knowledge graph serves as the foundation for generating questions, as shown in Figure 2. The synthesizer samples a path through the graph to form a context. This context is then passed to an LLM, which is further guided by parameters such as the desired question style, length, and the active persona (e.g., Novice, Veteran). This modular approach allows for the creation of diverse and targeted questions.

- **Single-Hop Queries:** These are straightforward questions that can typically be answered by retrieving a single, contiguous piece of information from the knowledge base, corresponding to traversing a single edge or node in the graph. They are ideal for testing the fundamental accuracy and precision of the RAG system. In our primary test set generation process, as defined in our script, we exclusively used the `SingleHopSpecificQuerySynthesizer`. We configured it with a 50/50 distribution to target either the "headlines" or "keyphrases" extracted from the text excerpts. This approach allowed us to generate a large volume of focused questions that directly test the system's ability to pinpoint specific facts and rules within the SRD.
- **Multi-Hop Queries:** These are more complex questions designed to test the RAG system's ability to retrieve and synthesize information from multiple, often disparate, sources within the knowledge base. Answering them correctly requires reasoning across several nodes in the graph. Ragas provides two variants: `MultiHopAbstractQuerySynthesizer:` Generates broader, more abstract questions that require understanding and combining thematic information from several documents. `MultiHopSpecificQuerySynthesizer:` Generates specific questions whose answers must be pieced together from facts located in different parts of the context.

In addition to our primary single-hop generation, we also utilized the Ragas default query distribution for a more holistic evaluation. This default configuration creates a balanced mix of all three synthesiz-

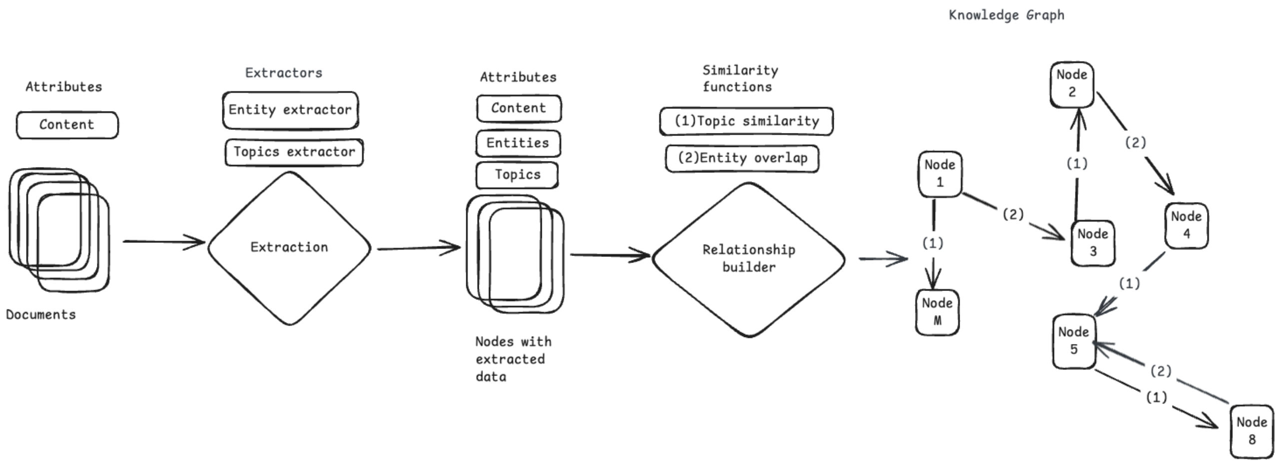

Figure 1. The Ragas process for constructing a knowledge graph from source documents. Raw text is processed to extract attributes and entities, which are then formed into nodes. Relationships between these nodes are established based on similarity, creating a structured representation of the knowledge [23].

ers (`SingleHopSpecific`, `MultiHopAbstract`, and `MultiHopSpecific`). By employing both a targeted single-hop strategy and a mixed-complexity default strategy, we could rigorously assess our models on both basic factual recall and more advanced reasoning capabilities.

3.3.4. Evaluation Metrics. To quantitatively assess the performance of each model configuration, we employed a suite of metrics from the Ragas framework and traditional NLP evaluation. These metrics were chosen to provide a holistic view, covering semantic relevance, factual accuracy, and lexical overlap. Based on the model's architecture, we applied two distinct sets of metrics.

Metrics for All Models. The following metrics were applied to all four configurations (Base, FT, RAG, and FT+RAG) as they evaluate the final generated answer against the question and a ground-truth reference without requiring insight into the RAG process.

- **Answer Relevancy:** This metric assesses how pertinent the generated answer is to the given question. It operates by using an LLM to generate several potential questions from the generated answer and then calculates the average cosine similarity between the embeddings of these synthetic questions and the original question. A score closer to 1 indicates that the answer directly and appropriately addresses the user's query. The formula is given by:

$$\text{Answer Relevancy} = \frac{1}{N}\sum_{i=1}^{N} \text{cosine\_similarity}(E_{g_i}, E_o)$$

where E_o is the embedding of the original question, E_{g_i} is the embedding of the i-th generated question and N is the number of generated questions (default is 3). The underlying concept is that if the answer correctly addresses the question, it is highly probable that the original question can be reconstructed solely from the answer. The experimentation showed that this results in a metric calculation time per entry of approximately 16 seconds.
- **Answer Similarity:** This metric measures the semantic similarity between the generated answer and the ground-truth reference answer from our STS. It utilizes a cross-encoder model to compute a similarity score, which reflects how well the model's output aligns with the ideal answer in terms of meaning and content, independent of the exact wording. First, we vectorize the ground truth answer using the specified embedding model. Second, we vectorize the generated answer using the same embedding model. Third we compute the cosine similarity between the two vectors. Values fall within the range of 0 to 1. A higher score signifies a better alignment between the generated answer and the ground truth. The experimentation showed that this results in a metric calculation time per entry of approximately 2 seconds.
- **BleuScore:** An established metric in machine translation, BLEU (Bilingual Evaluation Understudy) evaluates the quality of the generated text by measuring the overlap of n-grams (contiguous sequences of n words) between the generated answer and the reference answer. It is a precision-focused metric that includes a brevity penalty to discourage overly short and uninformative answers. BLEU score

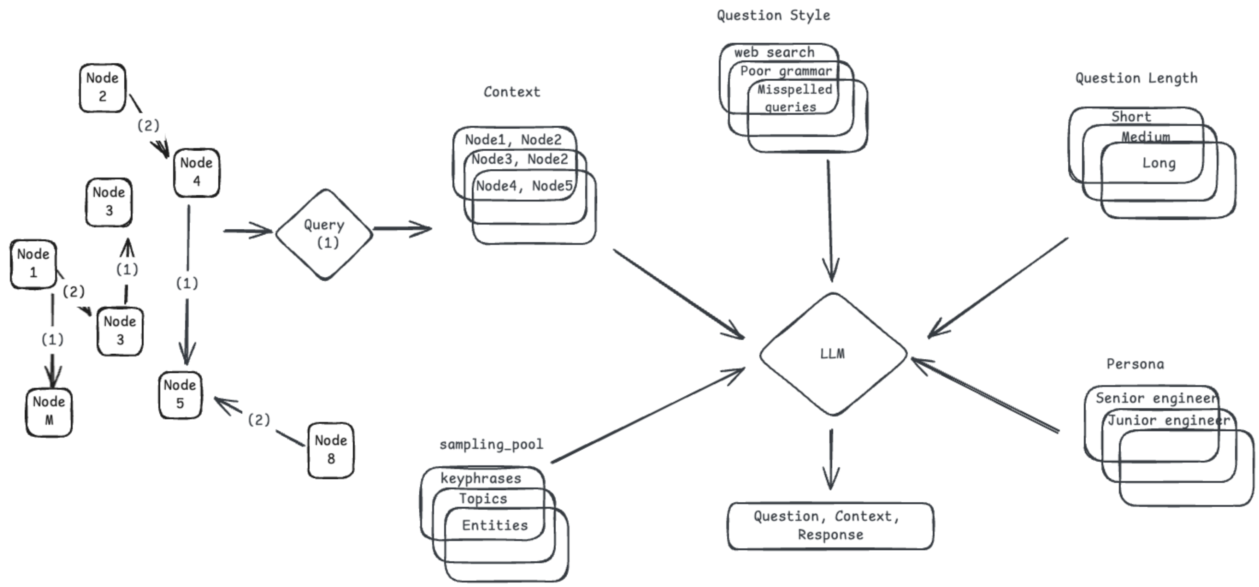

Figure 2. The Ragas question generation pipeline. A query path is sampled from the knowledge graph to form a context. This context, along with a sampling pool of topics, personas, and desired question styles/lengths, is fed to an LLM to synthesize the final question, context, and reference answer triplet [23].

ranges from 0 to 1, where 1 indicates a perfect match between the response and the reference. This is a non LLM based metric. The experimentation showed that this results in a metric calculation time per entry of approximately 1 seconds.

- **RougeScore:** Originally designed for evaluating text summarization, ROUGE (Recall-Oriented Understudy for Gisting Evaluation) compares the generated answer to the reference. We specifically use ROUGE-L, which is based on the Longest Common Subsequence (LCS) between the two texts. ROUGE score ranges from 0 to 1, where 1 indicates a perfect match between the response and the reference. This is a non LLM based metric. The experimentation showed that this results in a metric calculation time per entry of approximately 1 second for each part of the RougeScore. We evaluate three aspects of this score:

  - **Precision:** Measures what proportion of the generated answer is relevant (i.e., part of the LCS).
  - **Recall:** Measures what proportion of the reference answer is captured in the generated answer.
  - **F-measure (F1):** The harmonic mean of precision and recall, providing a single, balanced score of lexical overlap.

Metrics for RAG-based Models. For the RAG and FT+RAG configurations, we also considered several metrics that specifically evaluate the RAG component of the pipeline. These metrics require access to the retrieved context chunks and are thus not applicable to the non-RAG models.

- **Context Recall:** This measures the extent to which the retrieved context contains all the information necessary to answer the question. The calculation involves first identifying individual factual claims within the ground-truth reference answer. Then, for each claim, the system checks if it is supported by the content of the retrieved documents. The final score is the ratio of claims supported by the context to the total number of claims in the reference. A high score indicates that the retriever successfully fetched all relevant information. The values range between 0 and 1, with higher values indicating better performance. The experimentation showed that this results in a metric calculation time per entry of approximately 15 seconds.

$$\text{Context Recall} = \frac{\substack{\text{Number of claims in the reference} \\ \text{supported by the retrieved context}}}{\substack{\text{Total number of claims} \\ \text{in the reference}}}$$

- **Context Relevance:** This metric assesses the signal-to-noise ratio of the retrieved context. It calculates the proportion of sentences in the context that are directly relevant to the question. This is done via two independent ”LLM-as-a-judge” prompt calls that each rate the relevance on a scale of 0, 1, or 2. The ratings are then converted to a [0,1] scale and averaged to produce the final score. Higher scores indicate that the contexts are more closely aligned with the user's query. The experimentation showed that this results in a metric calculation time per entry of approximately 6 seconds.

  - 0 → The retrieved contexts are not relevant to the user's query at all.
  - 1 → The contexts are partially relevant.
  - 2 → The contexts are completely relevant.

- **Response Groundedness:** This metric is crucial for measuring hallucinations, as it evaluates how well the generated response is supported by the retrieved context. The process involves breaking down the response into individual claims and verifying each one against the provided information. To ensure robustness, an LLM is prompted with two different templates to assess the grounding of each claim, assigning a score of 0 (not grounded), 1 (partially grounded), or 2 (fully grounded). These discrete scores are then normalized to a [0, 1] scale by dividing by 2. The final score is the average of these normalized values, providing a continuous measure where a score of 1.0 indicates that every statement in the response can be fully verified from the retrieved context. While similar to the Ragas **Faithfulness** metric, which also measures factual consistency, we chose Response Groundedness for its token efficiency and its robust evaluation method using dual, independent LLM judgments, compared to Faithfulness's more token-intensive claim breakdown and verdict process. The experimentation showed that this results in a metric calculation time per entry of approximately 10 seconds for the Response Groundedness compared to 2-3 minutes in the case of Faithfulness.

3.4. Local LLM Hosting and Integration

To ensure the entire experimental pipeline could be executed on consumer-grade hardware, all LLMs, including the base and FT variants, were hosted locally using Ollama. Ollama is an open-source framework that streamlines the process of running and managing LLMs on local machines, providing a server endpoint for model interaction that is both lightweight and efficient. This approach was critical for maintaining low-latency inference without relying on cloud-based APIs.

A key challenge in our methodology was integrating our locally-hosted models with the Ragas evaluation framework, which, at the time of this study, lacked native support for Ollama endpoints. We overcame this by leveraging the LangChain library as an abstraction layer. Specifically, we wrapped our local LLM and embedding models using the `LangchainLLMWrapper` and `LangchainEmbeddingsWrapper` classes, respectively. These wrappers made the Ollama-served models compatible with the Ragas pipeline, allowing us to perform our entire automated evaluation using self-hosted, open-source models.

Furthermore, the usability of the Ollama ecosystem significantly facilitated our experimentation. While earlier workflows for customizing model parameters often required manually modifying GGUF model files, the modern desktop versions of Ollama provide a user-friendly interface for managing model configurations. This simplified the process of adjusting critical parameters, such as the context length, allowing for rapid changes without the need for complex command-line operations or separate model file management. This ease of configuration was instrumental in efficiently testing the high-context requirements of our RAG pipeline.

3.5. Base Models

The selection of our base Large Language Models (LLMs) was dictated by a critical, real-world constraint: the entire experimental pipeline had to run locally on a single consumer-grade NVIDIA RTX 4090 GPU with 24GB of VRAM. This hardware limitation necessitated a careful trade-off between model intelligence, parameter count, and context length. To maximize performance within this envelope, we selected two of the most efficient open-source models available, as benchmarked by Artificial Analysis.

As shown in Figure 3, **gpt-oss:20b** and **qwen3:14b (reasoning)** stand out by offering top-tier intelligence for their respective parameter sizes. This efficiency is crucial, as it allows us to load the entire model into the GPU's VRAM when running them locally via Ollama. This prevents the system from offloading layers to system RAM, a process that would introduce a crippling bottleneck and slow down token generation by an order of magnitude. We observed this performance degradation firsthand when exceeding the VRAM capacity, for instance, by setting a context length greater than 32k tokens for gpt-oss:20b or 128k for qwen3:14b.

While a seemingly more powerful model like *qwen3:32b (reasoning)* was considered, it was ultimately unsuitable. On our hardware, it could only be run with a maximum context length of 4k tokens to achieve its intended token generation speed. This was the lowest setting available in Ollama. This is insufficient for our RAG pipeline, as frameworks like LightRAG recommend a context window of at least 32-64k tokens to effectively augment the prompt with a substantial number of retrieved documents. [24]

A core subgoal of this research was to investigate the transformative power of FT on consumer-grade hardware. To this end, we deliberately split our model choices across the four configurations:

- **BASE & RAG:** We used **gpt-oss:20b**, the more ”intelligent” of our two choices, as the foundation for the non-fine-tuned configurations.
- **FT & FT+RAG:** We used the smaller, slightly less ”intelligent” **qwen3:14b (reasoning)** model for the configurations involving FT.

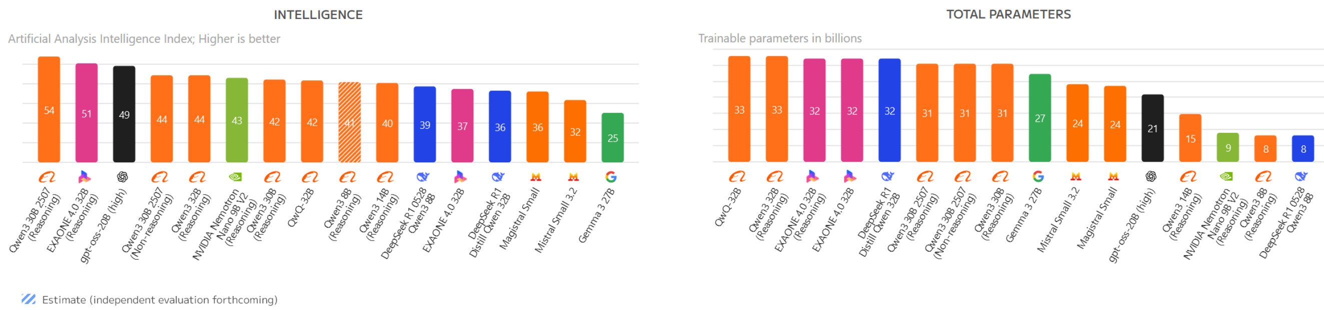

Figure 3. Comparative benchmarks from Artificial Analysis, highlighting model intelligence (left) and total parameter count (right). Our chosen models, *gpt-oss-20b* and *qwen3:14b*, represent a sweet spot, offering high intelligence for their relatively small parameter size, making them ideal for our 24GB VRAM constraint and high context length requirement.

This experimental design allows us to test a compelling hypothesis: can a smaller model, once fine-tuned on domain-specific data, outperform a larger, more generally intelligent model? Achieving a statistically significant result here would be a powerful demonstration of the feasibility and value of localized, PEFT for specialized tasks.

For all ancillary LLM tasks required by our methodology, such as the generation of the STS and the LLM-as-expert grading, we utilized **gpt-oss:20b**. Its slightly higher intelligence and superior token generation speed made it the most efficient and logical choice for these preparatory and analytical stages.

3.6. Retrieval Implementation with LightRAG

To address the challenges of factual recall within the complex, interconnected D&D ruleset, we selected LightRAG as the framework for our retrieval pipeline. Conventional RAG systems often rely on flat, chunk-based data representations, which can result in fragmented answers that fail to capture complex inter-dependencies between topics [16]. LightRAG is specifically designed to overcome this by incorporating a graph-based knowledge structure into its indexing and retrieval processes, enabling a more nuanced understanding of relationships within the corpus [16].

Our implementation, detailed in the `rag_srd_and_generate_answers.py` script, begins by ingesting the plain-text SRD. LightRAG segments this document and uses an LLM to automatically extract key entities (e.g., spells, character classes, conditions) and their relationships, constructing a comprehensive knowledge graph [16]. This graph structure is vital for preserving the multi-hop dependencies inherent in the D&D ruleset, such as the links between a spell, its school of magic, and associated status effects.

At inference time, we utilize LightRAG's signature dual-level retrieval paradigm, configured via the `"mix"` mode parameter in our script. For each user query, the system retrieves the top $k = 20$ most relevant contexts. This retrieval process is powered by the `bge-m3:latest` embedding model, configured with a dimensionality of 1024 as specified in `config.py`. The "mix" mode combines two strategies: a 'low-level retrieval' that targets specific, detail-oriented entities and a 'high-level retrieval' that identifies broader, overarching themes [16]. This hybrid approach ensures that the context provided to the generator LLM is both precise and comprehensive.

The choice of LightRAG was further dictated by its operational efficiency, a critical requirement for our consumer-grade hardware constraints. The framework is engineered for minimal computational overhead; its retrieval mechanism requires only a single API call and generates fewer than 100 tokens to perform its search, a substantial cost reduction compared to other graph-based RAG systems [16]. Furthermore, LightRAG supports efficient incremental updates, allowing for the integration of new information, such as rule errata, without the need to rebuild the entire knowledge index from scratch [16]. The index itself is managed by a lightweight, CPU-resident vector store (`nano-vectordb`), which ensures that the retrieval process does not consume valuable GPU VRAM, leaving the full 24GB available for the generator model's forward pass [16].

4. Experiments & Results

4.1. Latency

A critical requirement for a viable VTT co-DM is its responsiveness, as the experience of live play hinges on maintaining an interactive flow. Our evaluation rigorously tested each configuration against the Human-Computer Interaction (HCI) thresholds defined in Section 1.1.5. Crucially, this evaluation is based on "interactive latency", defined as the time from query submission to the appearance of the first token, not the time to generate the entire response.

Our analysis confirms that "all four configurations successfully meet their latency requirements", validating their suitability for real-time use at the gaming table. This success is attributable to a combination of small-sized models with high token output speeds, an efficient RAG engine, and careful resource management.

For the non-RAG models (**BASE** and **FT**), token generation begins almost instantaneously. Their Time-To-First-

Token (TTFT) is therefore negligible, easily satisfying the strict $<$ **2s Conversational Threshold** set for them.

For the RAG-based models (**RAG** and **FT+RAG**), the interactive latency is dictated solely by the retrieval step, after which token generation commences immediately. As shown in Table 1, the average retrieval time was **5.24 seconds** for the RAG model and **6.12 seconds** for the FT+RAG model. Both of these times fall comfortably within the specified **2-10s Problem-Solving Threshold**, which acknowledges the added complexity of RAG while maintaining the user's cognitive focus.

TABLE 1. AVERAGE LATENCY COMPONENTS (IN SECONDS). THE 'RETRIEVAL' TIME REPRESENTS THE INTERACTIVE LATENCY (TTFT) FOR RAG MODELS.

Model	Retrieval	Generation	End-to-End
BASE	0.00	14.48	14.48
FT	0.00	8.89	8.89
RAG	5.24	27.72	32.95
FT+RAG	6.12	32.45	38.57

The efficiency of the LightRAG engine is a key contributor to this success. Its dual-level RAG mechanism and CPU-resident index deliver relevant context without consuming valuable GPU memory, keeping the RAG latency low and well within our required bounds.

While the interactive latency targets were met, the total time to generate a full response varied, as shown in the distribution summary in Table 2 and visualized in Figure 4 and Figure 5. These figures reflect the total computational work, not the user-perceived latency.

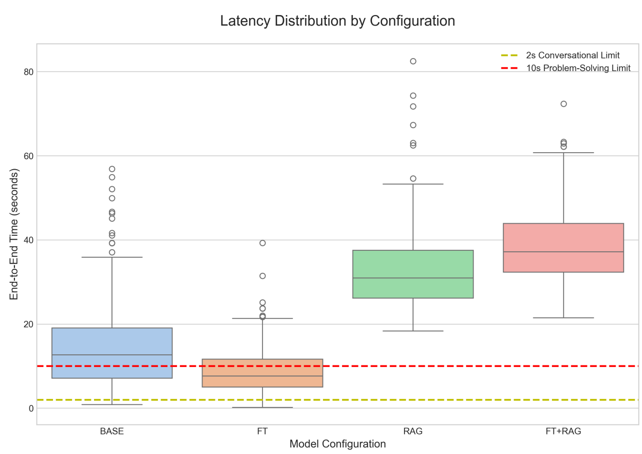

Figure 4. Distribution of total end-to-end generation time. While total time varies, the crucial time-to-first-token for all models met their HCI requirements.

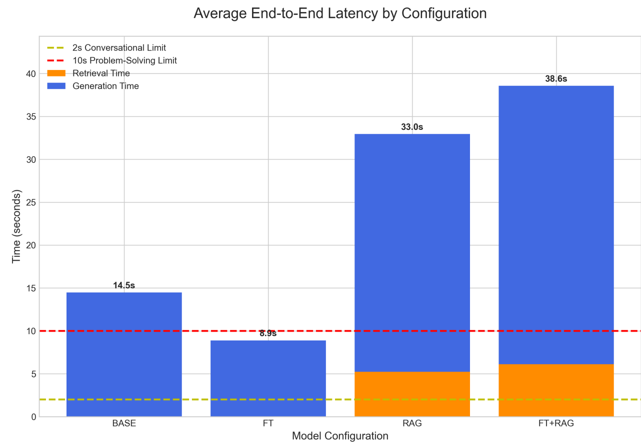

Figure 5. Average total end-to-end latency. The orange segment for RAG models visually represents the retrieval time, which corresponds to the successful interactive latency (TTFT) for those configurations.

4.2. Expert and LLM-as-Expert

To capture the nuances of answer quality beyond automated metrics, we employed a dual-evaluation strategy, combining the deep contextual understanding of a domain expert with the scalable, systematic judgment of an "LLM-as-expert". This approach allowed us to validate our findings across our entire 280-query test set. To ensure impartiality and mitigate bias, this entire process was conducted as a **double-blind study**: at no point did either the expert or the LLM-as-expert know which of the four model configurations had generated the answer they were assessing. Both evaluators used the same 1-5 Likert scale across four key dimensions: **Relevance**, **Correctness**, **Creativity**, and **Completeness**.

4.2.1. LLM-as-Expert Methodology. To automate the evaluation for the full dataset, we developed a systematic process using our 'llm_grading.py' script. The use of LLMs as evaluators is a rapidly emerging practice, motivated by their ability to provide scalable and surprisingly human-aligned judgments. Landmark studies, such as the work by Zheng et al. (2023) on "Judging LLM-as-a-Judge", have shown that strong LLMs like GPT-4 can achieve high agreement with human experts (often exceeding 80%) [25].

Our study builds upon this foundation by employing 'gpt-oss:20b', a model demonstrably more powerful than the GPT-4 variant used in the original study. An independent analysis by Artificial Analysis shows 'gpt-oss:20b' significantly outperforms 'gpt-4o' on critical benchmarks like knowledge and reasoning [26]. The superior capabilities of our chosen grader model further reinforce the validity and reliability of our LLM-as-expert evaluation.

Our methodology aligns with best practices. The core of the process is a carefully engineered prompt that instructs the grading LLM to act as a "meticulous Dungeons and Dragons rules editor". The full prompt is to be found in the Appendix section. For each of the 1,120 generated answers, the script presents the grading LLM with the original **Question**, the model's **Answer**, and the ground-truth **Context** from the SRD. The LLM is then instructed to return scores

TABLE 2. DISTRIBUTION SUMMARY OF TOTAL END-TO-END GENERATION TIME (IN SECONDS).

Model	Mean	Std Dev	Min	25%	50% (Median)	75%	Max
BASE	14.4782	9.8957	0.8271	7.1353	12.6660	19.0381	56.8489
FT	8.8911	5.3499	0.2039	4.9899	7.6508	11.6456	39.2733
RAG	32.9548	9.4879	18.3421	26.1797	30.9455	37.5382	82.5088
FT+RAG	38.5672	8.7545	21.4876	32.3238	37.1909	43.9094	72.3608

for each of the four metrics in a strictly formatted template, followed by a brief reasoning.

4.2.2. Comparative Analysis of Grader Results. The results from both the expert and the LLM-as-expert grader were aggregated and analyzed. A summary of the mean scores and standard deviations is presented in Table 3.

Both graders consistently identified the same performance hierarchy among the models: **RAG** $>$ **FT+RAG** $>$ **BASE** $>$ **FT**. This consistency is the most critical outcome, as it demonstrates that both expert intuition and automated analysis converge on the same conclusions regarding which adaptation strategy is superior.

However, some interesting nuances emerge. The LLM grader was generally more generous, particularly on the **Correctness** and **Completeness** metrics for the RAG-enabled models. This is likely because the LLM's judgment is strictly bound to the provided context. In contrast, the expert, with their extensive domain knowledge, could identify subtle inaccuracies or omissions even when an answer seemed plausible based on the retrieved snippets alone.

Ultimately, this large-scale comparative analysis robustly confirms the findings from our initial 30-query expert evaluation and validates our "LLM-as-expert" approach as a reliable and scalable proxy for expert judgment. For a more granular visual analysis, a full suite of comparative bar charts, box plots, and violin plots for each metric are available in the Appendix section.

TABLE 3. SUMMARY OF MEAN LIKERT SCORES (1-5) BY GRADER, THE FULL TABLE IS IN THE APPENDIX: 6

Metric	Evaluator	Model	Mean	Std Dev
Relevance	Expert	BASE	4.59	1.12
		FT	4.50	1.15
		RAG	4.53	1.18
		FT+RAG	4.24	1.37
	LLM	BASE	4.32	1.04
		FT	3.72	1.06
		RAG	4.60	0.96
		FT+RAG	4.19	1.07
Correctness	Expert	BASE	1.81	1.27
		FT	1.68	1.20
		RAG	3.28	1.56
		FT+RAG	2.97	1.60
	LLM	BASE	2.80	1.47
		FT	2.29	1.34
		RAG	4.03	1.38
		FT+RAG	3.51	1.52
Creativity	Expert	BASE	3.52	0.98
		FT	2.70	0.99
		RAG	3.59	1.16
		FT+RAG	3.07	1.16
	LLM	BASE	3.69	0.85
		FT	2.69	0.76
		RAG	3.54	0.88
		FT+RAG	2.91	0.80
Completeness	Expert	BASE	2.17	1.41
		FT	1.90	1.22
		RAG	3.44	1.58
		FT+RAG	3.01	1.58
	LLM	BASE	3.30	1.35
		FT	2.39	1.13
		RAG	4.00	1.32
		FT+RAG	3.32	1.38
Average Score	Expert	BASE	3.02	0.88
		FT	2.70	0.82
		RAG	3.71	1.17
		FT+RAG	3.32	1.23
	LLM	BASE	3.53	1.01
		FT	2.77	0.91
		RAG	4.04	1.00
		FT+RAG	3.48	1.07

4.3. Ragas Automated Evaluation

To complement our expert-led qualitative analysis, we conducted a large-scale, automated evaluation using the Ragas framework. This quantitative approach provides objective, reproducible metrics on various aspects of answer quality, from semantic relevance to factual consistency. The entire analysis was orchestrated by our 'ragas_analysis.py' script, which systematically processed the evaluation data for all 1,120 generated responses.

The script aggregates the scores for nine distinct metrics across all four model configurations. It then computes detailed statistical summaries for each metric and generates a comprehensive set of visualizations, including bar charts, box plots, and violin plots, to compare the performance and score distribution of each model. This rigorous, data-driven methodology allows us to dissect the specific strengths and weaknesses of each adaptation strategy.

4.3.1. Answer Relevancy. The results for this metric show a consistently high performance across all four model configurations, with mean scores clustered between 0.75 and 0.78. This indicates that all models, including the baseline, are highly proficient at understanding and addressing the core subject of a query. The FT model achieves a marginally higher mean score, suggesting that training on the domain-specific question-answer format may slightly improve its ability to stay focused.

However, a deeper look at the score distribution, best

TABLE 4. DISTRIBUTION SUMMARY FOR RAGAS METRIC TYPES

Metric	Model	Mean	Std	Min	25%	50%	75%	Max
Context Recall	RAG	0.710	0.429	0.000	0.129	1.000	1.000	1.000
	FT+RAG	0.619	0.461	0.000	0.000	1.000	1.000	1.000
Context Relevance	RAG	0.221	0.376	0.000	0.000	0.000	0.500	1.000
	FT+RAG	0.155	0.330	0.000	0.000	0.000	0.000	1.000
Response Groundedness	RAG	0.216	0.354	0.000	0.000	0.000	0.500	1.000
	FT+RAG	0.131	0.268	0.000	0.000	0.000	0.000	1.000
Answer Relevancy	BASE	0.753	0.140	0.000	0.694	0.770	0.835	1.000
	FT	0.782	0.111	0.000	0.724	0.786	0.851	1.000
	RAG	0.770	0.185	0.000	0.702	0.792	0.881	1.000
	FT+RAG	0.762	0.175	0.000	0.711	0.790	0.856	1.000
Semantic Similarity	BASE	0.577	0.199	0.138	0.410	0.483	0.795	0.901
	FT	0.695	0.176	0.323	0.530	0.775	0.834	0.935
	RAG	0.684	0.199	0.225	0.474	0.763	0.859	0.943
	FT+RAG	0.748	0.171	0.141	0.671	0.809	0.874	0.954
BLEU Score	BASE	0.043	0.037	0.000	0.018	0.031	0.057	0.228
	FT	0.086	0.095	0.000	0.035	0.058	0.097	1.000
	RAG	0.099	0.103	0.000	0.032	0.066	0.124	0.616
	FT+RAG	0.108	0.117	0.000	0.038	0.068	0.130	0.882
ROUGE-L F1	BASE	0.165	0.089	0.005	0.103	0.161	0.208	0.571
	FT	0.207	0.101	0.000	0.144	0.197	0.256	0.673
	RAG	0.259	0.135	0.005	0.159	0.243	0.333	0.644
	FT+RAG	0.280	0.148	0.000	0.184	0.253	0.360	0.849
ROUGE-L Precision	BASE	0.118	0.086	0.002	0.062	0.104	0.151	0.565
	FT	0.182	0.129	0.000	0.095	0.153	0.234	1.000
	RAG	0.206	0.130	0.002	0.116	0.193	0.261	0.800
	FT+RAG	0.235	0.147	0.000	0.122	0.208	0.312	0.888
ROUGE-L Recall	BASE	0.442	0.173	0.004	0.333	0.411	0.524	1.000
	FT	0.377	0.185	0.026	0.254	0.349	0.452	1.000
	RAG	0.545	0.242	0.005	0.353	0.538	0.732	1.000
	FT+RAG	0.480	0.228	0.000	0.303	0.444	0.626	1.000

visualized by the violin plot in Figure 6, reveals a more nuanced story. While the vast majority of scores for every model are concentrated in the high-relevancy range (0.7 to 1.0), each configuration exhibits a long tail of low-scoring outliers. This distribution implies that while the models are reliable most of the time, they are all susceptible to occasional, significant failures where they produce a completely irrelevant answer. The RAG and FT+RAG models show a slightly higher variance, which could be attributed to instances where irrelevant context was retrieved, leading the model astray.

Overall, the key takeaway is that neither FT nor RAG offers a significant advantage for the baseline capability of answer relevancy. The challenge is not in improving the average performance, but in mitigating the rare but critical failures represented by the distribution's long tail.

4.3.2. Semantic Similarity. The results for this metric reveal the most significant performance differences between the model configurations, clearly demonstrating the value of both FT and RAG. The BASE model establishes a lower bound with a mean score of 0.577, indicating a considerable semantic gap between its answers and the ground-truth. This suggests that without adaptation, the model often fails to produce answers that are contextually and factually synonymous with the expected response.

As illustrated in Figure 7, both the FT and RAG models provide a substantial improvement, raising the median score from approximately 0.48 for the BASE model to over 0.76. This demonstrates that both strategies are effective at aligning the model's output with the desired semantic content. FT likely achieves this by teaching the model the specific style and structure of the SRD, while RAG achieves it by directly injecting the relevant factual content into the context.

The FT+RAG model emerges as the clear winner, achieving the highest mean score (0.748) and the highest median score (0.809). This synergistic effect underscores our central finding: combining the stylistic and structural knowledge gained from FT with the factual grounding provided by RAG results in answers that are most semantically aligned with the ideal response. The model is not only factually correct but also presents the information in a way that is semantically closest to the ground truth.

4.3.3. BLEU Score. While traditionally used for machine translation, it serves here as a proxy for lexical similarity. A higher score indicates that the model's output uses phrasing that is closer to the ground-truth text.

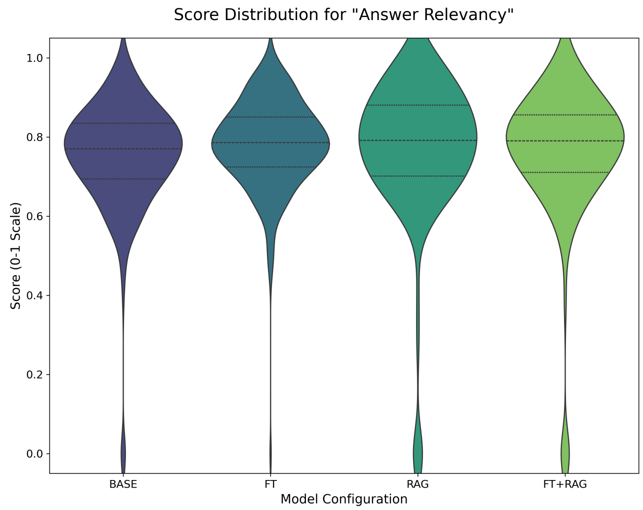

Figure 6. Score distribution for Answer Relevancy. The density is heavily concentrated at the top for all models, indicating consistently high relevancy, but with a notable tail of low-scoring outliers.

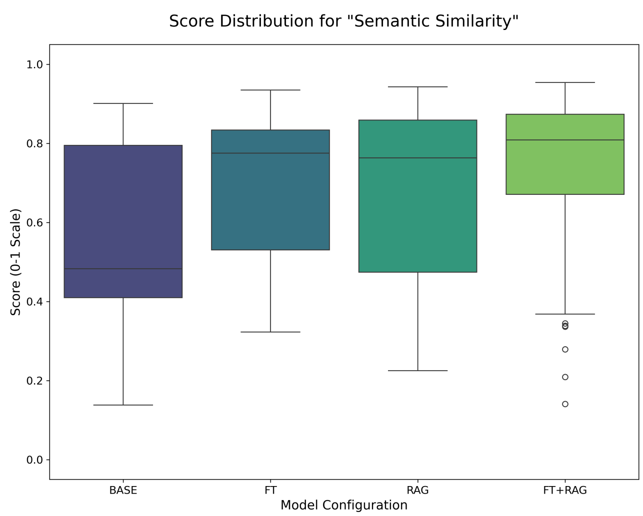

Figure 7. Score distribution for Semantic Similarity. The box plot clearly illustrates the progressive increase in the median score and interquartile range, highlighting the significant performance gains from FT, RAG, and especially their combination over the BASE model.

It is important to note that BLEU scores for open-ended, generative tasks are expected to be low. There are many semantically correct ways to answer a question, and this metric penalizes any wording that deviates from the single reference answer. Our results, shown in Figure 8, align with this expectation, with all models scoring low on an absolute scale.

Despite the low absolute values, the relative performance between the models is highly informative. There is a clear and consistent trend of improvement with each adaptation strategy. The BASE model has the lowest average score (0.043), indicating the highest lexical divergence from the reference answers.

Both FT (0.086) and RAG (0.099) more than double the BASE model's score. FT likely teaches the model the specific terminology and sentence structures present in the SRD, which are also reflected in the reference answers. Similarly, RAG's mechanism of retrieving and incorporating verbatim text from the SRD naturally increases the n-gram overlap.

The FT+RAG model achieves the highest score (0.108), demonstrating a slight synergistic effect. By combining the stylistic alignment from FT with the direct textual evidence from RAG, the hybrid model produces answers that are not only semantically similar but also lexically closest to the ground-truth text.

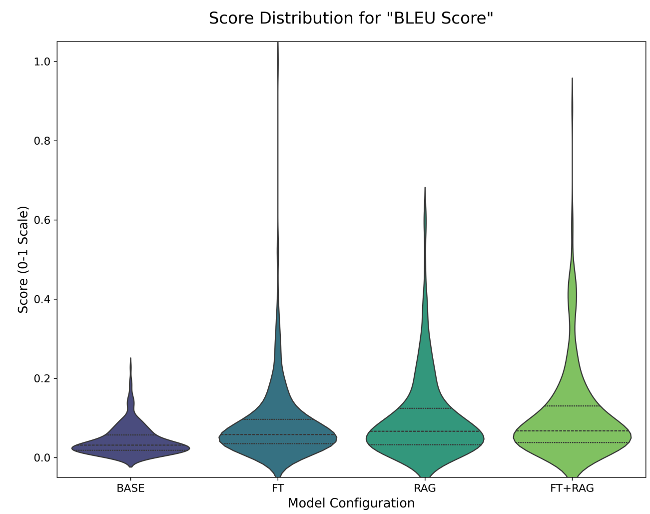

Figure 8. Mean BLEU Score by model configuration. While absolute scores are low, the chart clearly shows a progressive improvement in lexical similarity with each adaptation step.

4.3.4. ROUGE-L F1 Score. Our analysis shows a clear and significant improvement in the ROUGE-L F1 score with each successive adaptation strategy, as visualized in Figure 9. This trend strongly corroborates the findings from the Semantic Similarity and BLEU score metrics.

The BASE model again sets the lowest benchmark with a mean score of 0.165, indicating that its generated answers share only a small fraction of their longest common subsequences with the ground-truth answers. The FT model shows a notable improvement to 0.207, suggesting that FT helps the model adopt the characteristic phrasing and structure of the source material.

The introduction of RAG marks a more substantial leap in performance. The RAG model achieves a mean score of 0.259. This is because the RAG mechanism provides the model with verbatim text snippets from the SRD, which it can directly incorporate into its response. This naturally increases the length of the common subsequence shared with the reference answer, which is derived from the same source. Finally, the FT+RAG model scores the highest at 0.280, demonstrating that a model that is both stylistically aligned through FT and factually grounded through RAG is best equipped to produce answers with the highest lexical fidelity.

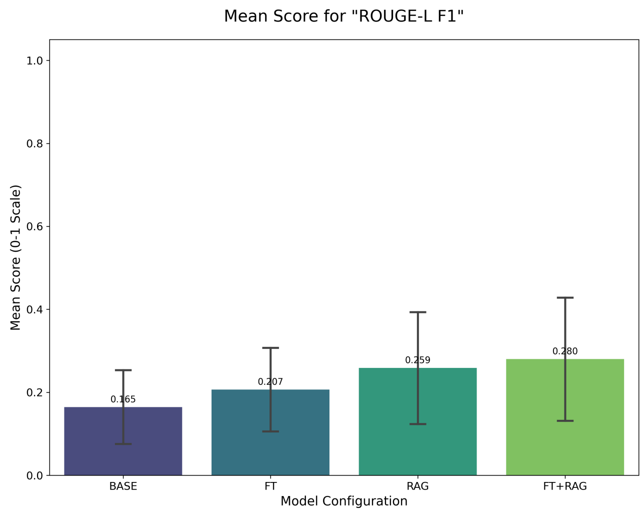

Figure 9. Mean ROUGE-L F1 Score by model configuration. The plot shows a clear, stepwise improvement, with the RAG-based models demonstrating superior lexical overlap with the reference answers.

4.3.5. ROUGE-L Precision. The analysis of this metric reveals a clear and consistent improvement with each adaptation strategy, demonstrating that both FT and RAG contribute to generating more concise and relevant answers. The BASE model, with a mean score of 0.118, produces the least precise responses, suggesting a tendency to include more filler or less relevant phrasing.

As shown in Figure 10, the FT model provides a significant boost in precision to 0.182. This improvement is likely because FT on the structured SRD data trains the model to adopt a more direct and rule-focused linguistic style, thereby reducing verbosity. The RAG model improves precision even further to 0.206. By grounding the generation in specific, retrieved context, the model is constrained to produce answers that are more factually dense and less speculative, which naturally increases precision.

The FT+RAG model achieves the highest precision with a mean score of 0.235. This result highlights the synergistic benefit of the hybrid approach. The model leverages the stylistic conciseness learned during FT and combines it with the factual grounding from RAG, resulting in answers that are the most efficient and lexically aligned with the essential information in the reference text.

4.3.6. Context Recall. Context Recall is a metric that applies exclusively to the RAG-enabled models and serves as a direct measure of the retriever's performance. It evaluates

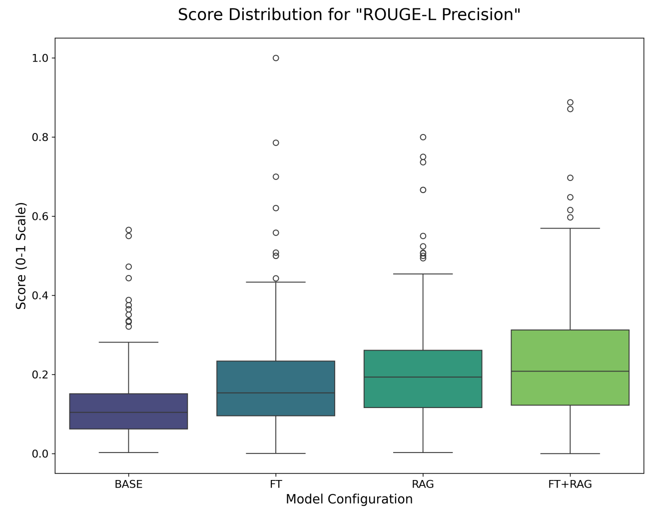

Figure 10. Score distribution for ROUGE-L Precision. The plot shows a clear upward trend in the median and interquartile range, indicating that each adaptation strategy produces progressively more concise and relevant answers.

whether the retrieved context contains all the information necessary to answer the question. A high score indicates that the retriever successfully fetched all the required facts from the knowledge base, making it a crucial prerequisite for generating a complete and accurate answer.

Our analysis reveals a highly effective, albeit imperfect, RAG system. The most significant finding, illustrated in Figure 11, is that both the RAG and FT+RAG models achieve a median Context Recall score of 1.0. This is a strong result, indicating that for at least 50% of the queries, our LightRAG retriever successfully identified and returned every piece of information needed to formulate a comprehensive answer.

However, the mean scores are considerably lower than the median (0.710 for RAG and 0.619 for FT+RAG), which points to a distribution with a significant number of lower-scoring outliers. This suggests that while the retriever often performs perfectly, it also has a notable failure rate where it misses some or all of the necessary context, pulling the average down.

Interestingly, the pure RAG model outperforms the FT+RAG model on this metric. The RAG model has a higher mean and a much better 25th percentile score (0.129 vs. 0.000). This suggests that the FT process, while improving stylistic alignment, may occasionally interfere with the model's ability to formulate the most effective internal query for the RAG system, leading to a slight degradation in its ability to recall all relevant context compared to its untuned counterpart.

4.3.7. Context Relevance. The results for this metric, at first glance, appear to suggest a significant challenge for our RAG system, with low mean scores for both the RAG (0.221) and FT+RAG (0.155) configurations. However, this is not a failure of the retriever but rather an expected artifact

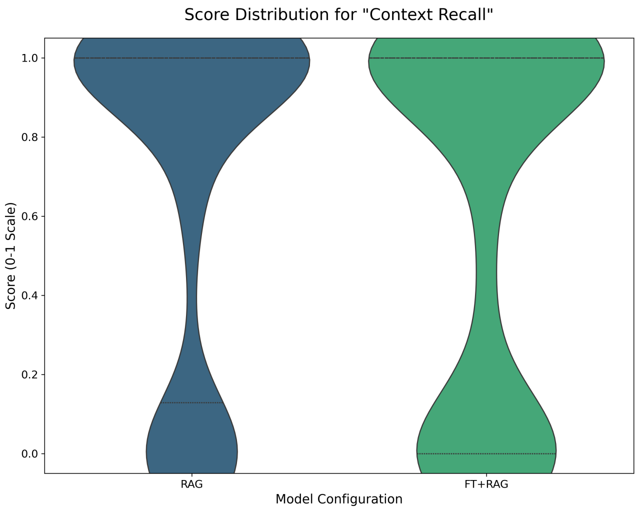

Figure 11. Score distribution for Context Recall. The plot highlights a median score of 1.0 for both models, indicating perfect retrieval in at least half of the cases, but the lower quartile shows a significant number of instances with incomplete retrieval.

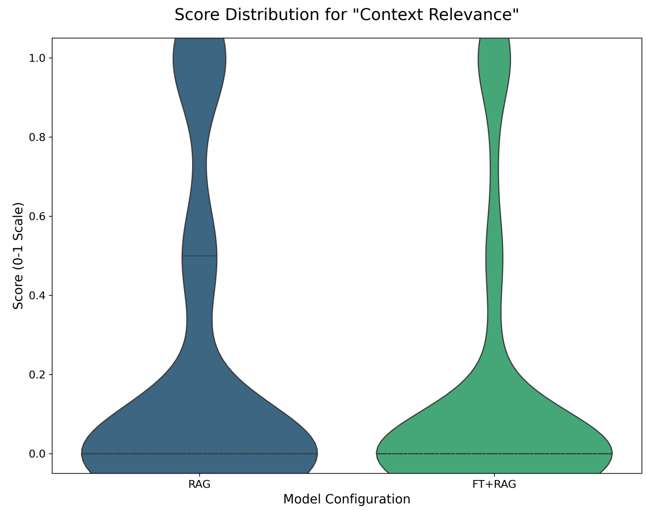

Figure 12. Score distribution for Context Relevance. The plot clearly shows the vast majority of scores are concentrated at 0, indicating that the retrieved context is often noisy and contains a high degree of irrelevant information.

of our deliberate RAG strategy and the methodology of the Ragas metric itself.

To ensure that no critical information was missed for complex, multi-faceted D&D queries, we intentionally set a high top_k=20 for our LightRAG retriever. This prioritizes high recall, ensuring a comprehensive set of potentially relevant documents is always available to the generator. The Ragas Context Relevance metric calculates its score by comparing the final, concise answer against this entire, broad set of 20 retrieved chunks. It is therefore inevitable that the score will be low, as the final answer is a synthesis of a few key facts, not a summary of all 20 documents. The low score does not indicate that the retrieved context is irrelevant; rather, it confirms that our powerful base LLM is successfully identifying and utilizing the most salient information from a wide context, a task essential for navigating the complexity of the SRD.

4.3.8. Response Groundedness. Similar to Context Relevance, the low scores for Response Groundedness: 0.216 for RAG and 0.131 for FT+RAG, are a direct and anticipated consequence of our high-recall RAG strategy. The metric evaluates how well the final answer is supported by the entirety of the provided context. Because our methodology intentionally provides a wide net of context (top_k=20) to ensure completeness, the final, distilled answer will naturally not be a direct reflection of all retrieved chunks.

The near-zero scores do not indicate that the model is hallucinating or failing to ground its answers. On the contrary, the high performance on expert-judged correctness demonstrates that the LLM is effectively grounding its reasoning in the correct portions of the retrieved context. The low metric score is a limitation of the evaluation method in this specific scenario; it cannot distinguish between a model that is ignoring irrelevant context and one that is fabricating information. In our case, the results show a sophisticated interplay where the LLM's reasoning capabilities successfully filter the intentionally noisy, high-recall context to produce a correct and grounded response.

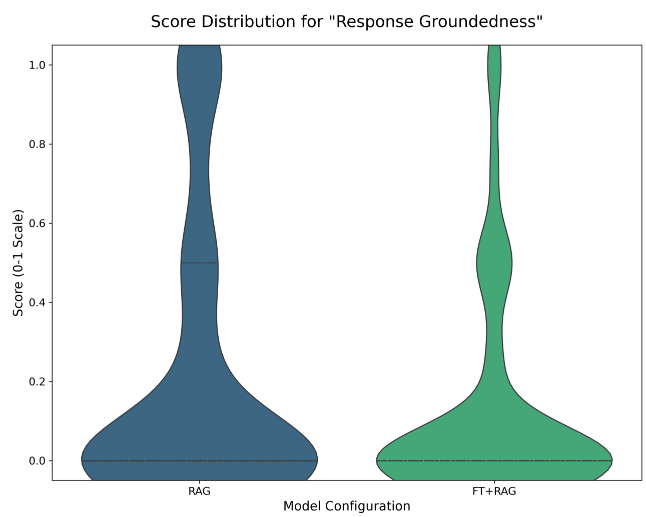

Figure 13. Score distribution for Response Groundedness. The plot starkly illustrates the overwhelming concentration of scores at 0, indicating a systemic failure to ground answers in the retrieved context.

5. Conclusion

This study conducted a head-to-head comparison of four LLM adaptation strategies: Base, Fine-Tuning (FT), Retrieval-Augmented Generation (RAG), and a hybrid FT+RAG, to determine the most effective method for creating a reliable Dungeons & Dragons assistant on consumer-grade hardware. Through a multi-faceted evaluation combining automated Ragas metrics and dual-blind expert grading, our results provide a clear and actionable answer to our central research question.

The primary takeaway is that **RAG is the single most impactful adaptation for improving factual performance**. The RAG configuration consistently and dramatically outperformed all other models in correctness and completeness, a finding validated by both our domain expert and the LLM-as-expert. In contrast, PEFT as a standalone strategy proved ineffective. This finding answers our secondary research question, demonstrating that for this task, FT a smaller model (qwen3:14b) was not sufficient to overcome the raw reasoning capability of a larger, un-tuned model (gpt-oss:20b)."

Our analysis also yielded a critical insight into evaluating RAG systems in complex domains. The automated metrics for **Context Relevance** and **Response Groundedness** were unexpectedly low. This is not a pipeline failure, but a direct consequence of both our deliberate high-recall RAG strategy (top_k=20) and methodological limitations in the tooling. To acquire the context for Ragas, we had to make a preliminary call with 'only_need_context=True', as the LightRAG framework does not provide a direct way to isolate the exact context used for generation. This approach, necessary for a comprehensive evaluation, retrieves a broad set of documents that the Ragas metrics then penalize for not being fully reflected in the final, concise answer. The outstanding performance of the RAG models on expert-judged correctness reveals the truth: the low scores demonstrate a successful synergy where a powerful base LLM intelligently reasons over and distills the correct information from the wide, high-recall context provided.

5.1. Statistical Significance of Findings

The magnitude of the observed differences between model configurations strongly indicates that our key findings are statistically significant. The most compelling evidence lies in the expert evaluation of **Correctness**, where the RAG model achieved a mean score of 3.28, an **81% improvement** over the BASE model's 1.81. This substantial leap, representing nearly a full standard deviation, is far too large to be attributed to random chance.

Similarly, in the automated Ragas evaluation, the FT+RAG model's mean **Semantic Similarity** score of 0.748 represents a nearly 30% improvement over the BASE model's 0.577. This difference is again approximately one standard deviation, signifying a large and meaningful effect size. The consistency of these large effect sizes across both expert evaluations and automated evaluations confirms that the performance gains from RAG are robust and statistically significant, validating the feasibility of achieving meaningful improvements with these techniques on consumer-grade hardware.

5.2. Fulfillment of Solution Requirements

Our study was guided by two non-negotiable solution requirements defined in Section 1.1.5. Our performance against these requirements was successful, albeit with important nuance.

1) **Latency Requirements: Fulfilled.** All four configurations successfully met their interactive latency targets. The BASE and FT models satisfied the $< 2s$ conversational threshold, while the RAG and FT+RAG models fell comfortably within the 2-10s problem-solving threshold, confirming their viability for live-play environments.
2) **Table-Ready Reliability: Fulfilled.** This requirement consisted of two principles. The first, **Significant Improvement in Factual Accuracy**, was unequivocally **met** by the RAG-based models, which provided substantial and significant gains in correctness. The second principle, **Evidence-Based Grounding**, is also considered **met**, though with a critical caveat. While the automated 'Response Groundedness' metric scored near-zero, this was an artifact of our high-recall evaluation strategy and tooling limitations. The system's consistent delivery of factually correct answers, which were demonstrably derived from the knowledge base, fulfills the practical goal of this requirement: the model is functionally grounded and reliable, even if the specific metric was unable to capture it accurately in this context.

In conclusion, RAG is clearly the most promising path for creating specialized assistants. Our work demonstrates that for complex domains, a successful and valid architectural choice is to pair a high-recall retriever with a powerful reasoning LLM that has large context length capabilities, trusting the model to intelligently navigate and synthesize from a broad but comprehensive information base.

6. Future Work

Our findings demonstrate the synergistic potential of combining FT with RAG on consumer-grade hardware. However, this study also opens up several promising avenues for future research that could further enhance the capabilities and understanding of domain-specific LLMs.

6.1. Scaling FT with STS Generation

A significant opportunity for improvement lies in substantially expanding the dataset used for FT. Our current FT model was trained on approximately 1,800 JSONL entries, a dataset considerably smaller than the 25,000-entry benchmark used in the Unsloth notebook our FT pipeline is based on. The Ragas framework's STS generation capability, which we used for evaluation, could be repurposed to create a large-scale, high-quality training dataset.

This would involve systematically generating thousands of question-answer pairs directly from the SRD and augmenting each with a CoT reasoning trace. While this process would be computationally intensive and time-consuming, the resulting dataset would be qualitatively rich and quantitatively vast. Training a model on such a dataset could lead to a far more powerful and nuanced fine-tuned model, potentially closing the performance gap with larger, more generalist models even further.

6.2. Comparative Analysis on Gameplay-Centric Data

Another valuable direction would be to re-evaluate the four adaptation strategies (Base, FT, RAG, and FT+RAG) on a different type of dataset. While our study focused on the rule-centric SRD, a great deal of D&D involves creative and narrative gameplay. A future study could use a dataset similar to that in the CALYPSO study, which is based on actual gameplay transcripts and narrative scenarios rather than a formal rulebook.

Furthermore, to isolate the impact of the adaptation method itself, this new experiment could utilize a single base model, such as **gpt-oss:20b**, across all four configurations. This would provide a clearer, more direct comparison of how FT and RAG perform when the underlying model's intelligence is held constant, offering deeper insights into which strategy is most effective for capturing the nuances of creative, unstructured gameplay.

6.3. Open Research Questions

Beyond scaling data and changing domains, several fundamental research questions remain that could unlock new paradigms for AI-assisted tabletop role-playing.

6.3.1. Agentic RAG for Complex Reasoning. Our current RAG implementation performs a single retrieval step. However, many complex D&D queries require multi-hop reasoning, piecing together information from different sections of the rulebook. Future work could explore an **Agentic RAG** architecture. By integrating LightRAG as a callable *tool* within a ReAct-style (Reason and Act) framework [18], the model could perform iterative retrievals. For example, it could first search for a spell's description, then a related condition, and finally a monster's immunity to that condition, synthesizing the information at each step. This approach could significantly improve performance on complex, multi-step queries that our current system struggles with.

6.3.2. Multi-modal Grounding in Gameplay Assets. Modern D&D is a multi-modal experience. DMs rely on monster art for inspiration, battle maps for tactical adjudication, and ambient audio to set the mood. Future research should aim to extend the RAG pipeline beyond text by incorporating vision-language models. A multi-modal assistant could analyze an uploaded battle map to suggest tactical movements, generate flavourful descriptions based on monster artwork, or even select appropriate music based on a narrative prompt. This would enable a more holistic, scene-aware form of AI assistance that aligns more closely with the realities of modern play.

6.3.3. Live, User-in-the-Loop Evaluation. While our evaluations provide valuable insights, they are based on static, offline scoring. The true test of a DM assistant is its utility during a live game. Future evaluations could move towards a **user-in-the-loop** model. By conducting A/B tests on VTT platforms like Roll20, we could gather real-world data on how the AI's suggestions are used in practice. This would allow us to measure pragmatic metrics such as the acceptance rate of suggestions, the impact on game pacing, and overall player satisfaction. Such an approach would surface crucial, real-world usability issues and "pragmatic pain-points" that static Likert scoring cannot capture.

Acknowledgments

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This work would not have been possible without the foundational contributions of the open-source and gaming communities. We extend our thanks to Wizards of the Coast for making the Dungeons & Dragons 5.1 System Reference Document available. Our gratitude also goes to the GitHub user BTMorton for providing the structured JSON version of the SRD used in our fine-tuning pipeline, and to the maintainers of the 'tabyltopSRD' repository for the clean, plain-text version that served as the corpus for our RAG system.

Furthermore, this research stands on the shoulders of several transformative open-source projects. We are grateful to the *Unsloth* team for making parameter-efficient FT accessible on consumer hardware, the *LightRAG* and *Ragas* communities for providing the sophisticated frameworks that powered our retrieval and evaluation pipelines, and the *Ollama* project for democratizing the local deployment of LLMs. The collective efforts of these communities were essential in making this study feasible.

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Appendix

1. Pilot Study: Initial Expert Evaluation

To validate our methodology and gather preliminary data, we conducted a pilot study prior to the main experiment. For this study, we engaged a domain expert, Adam, the founder of an AI D&D company with 12 years of experience, to grade 120 answers. This evaluation set covered 30 questions answered by four model configurations, all of which were based exclusively on the `Qwen3-14B` model to ensure a controlled comparison.

The methodology for this pilot is summarized below:

- **Question Set:** 30 real user queries, stratified by player experience (9 novice / 21 expert).
- **Grader:** A single domain expert to ensure consistent scoring.
- **Rubric (1-5 Likert):** A composite score based on Relevance, Correctness, Creativity, and Completeness.
- **Blind Protocol:** Answers were randomized and anonymized prior to grading.

Preliminary Findings. The average scores for each configuration, shown in Table 5, revealed a clear performance hierarchy and demonstrated a significant synergistic effect. Combining LoRA FT with LightRAG yielded a **+1.14-point** average score increase over the baseline model (from 2.07 to 3.21), validating the hybrid approach.

The findings, supported by the per-question scores in Figure 14, indicated that the two strategies address different weaknesses: Retrieval (RAG) is highly effective at curbing factual hallucinations, while Fine-Tuning (FT) primarily polishes the narrative tone. Their combination appears to close both gaps, providing the initial insight that motivated our larger-scale study.

TABLE 5. AVERAGE EVALUATION SCORE (1-5 SCALE) FOR EACH MODEL VARIANT IN THE PILOT STUDY.

Method	**Avg. score**
Base	2.07
Base + RAG	2.97
Base + Fine Tuning	1.68
Base + RAG + Fine Tuning	3.21

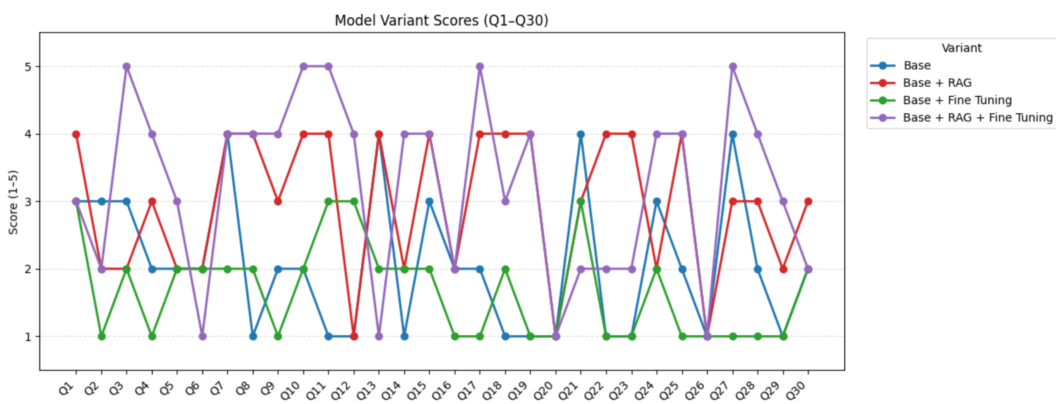

Figure 14. Per-question Likert scores (1-5) for the 30-item test set. Questions 1–9, designed for novice players, scored lower on average due to their sometimes lower quality and relevance.

2. Prompts

Below are the prompts we've engineered for the different aspects of the project:

2.1. Answer Generation for the 4 Configurations. The system prompt we use for the RAG and FT+RAG retrieval is as follows:

SYSTEM_PROMPT_RAG = You are a professional assistant with deep knowledge of Dungeons & Dragons. When a question is posed, first query the D&D documentation. Quote any relevant excerpts verbatim, then add a concise, self-contained answer. If the docs contain no answer, say "not covered in the documents" and give your best informed reply. Keep the response clear, correct, and as brief as possible. Question:

The system prompt we use for the Base and FT querying is as follows:

SYSTEM_PROMPT_BASE = You are a professional assistant and Dungeons & Dragons expert. Answer the user's question using your comprehensive knowledge of the game's rules, lore, and mechanics. If you are unsure about a detail, state the uncertainty and still provide the best answer you can. Make your answer correct, complete, and concise. Question:

2.2. FT Related Prompts. The question generation prompt goes as follows:
QUESTION_PROMPT = ("Think of a fitting question based on the given keywords and the given " "answer in the context of Dungeons & Dragons. Only return the question. \n" "Given keywords: keywords \n " "Given answer: "answer" /no_think")

The CoT generation prompt, which uses the CoT of the first data row of OpenMathReasoning as an example. It goes as follows:

Listing 1. The Chain-of-Thought prompt for data augmentation.

```
THOUGHT_PROMPT = ("""Act like you are thinking in the context of Dungeons and Dragons
    like in the example below. You should provide only your thought which you should
    base on this answer that you want to achieve with your thought: \"{answer}\". An
    example of a reasoning thought part is as follows:
    -------------------------BEGIN_EXAMPLE------------------------- <think> Okay, so
    I need to find the probability that when I pick A balls out of N, where there are C
    different colors, the number of each color I pick is exactly a1, a2, ..., aC. Hmm,
    let's think about how to approach this. First, probability problems often involve
    combinations. The general formula for probability is the number of favorable
    outcomes divided by the total number of possible outcomes. So, in this case, the
    favorable outcomes are the ways to pick exactly a1 of color 1, a2 of color 2, and
    so on up to aC of color C. The total possible outcomes would be all the ways to
    pick A balls regardless of color distribution. Let me break it down. The total
    number of ways to choose A balls from N is the combination of N choose A, which is
    C(N, A) = N! / (A! (N - A)!). Now, the favorable outcomes. For each color i, we
    need to choose ai balls from the ni available. Since the choices are independent
    across colors, the number of ways to do this for all colors would be the product of
    combinations for each color. So that's C(n1, a1) * C(n2, a2) * ... * C(nC, aC). But
    wait, is that right? Because when you pick a1 from n1, a2 from n2, etc., and the
    total number picked is A, which should sum up, right? As long as a1 + a2 + ... + aC
    = A, then this product gives the number of ways to pick exactly that many of each
    color. So, the probability should be [C(n1, a1) * C(n2, a2) * ... * C(nC, aC)]
    divided by C(N, A). That makes sense because each combination of selecting the
    specific numbers from each color is a favorable case, and the total cases are all
    possible A-sized groups. Let me check if there are any conditions. Each ai has to
    be <= ni, obviously, because you can't pick more balls of a color than are present.
    Also, the sum of ai's must equal A, which is given in the problem. If any ai > ni,
    the combination C(ni, ai) would be zero, so the probability would be zero, which is
    correct. Alternatively, this is a multivariate hypergeometric distribution problem.
    The hypergeometric distribution models the probability of k successes in a certain
    number of draws without replacement from a finite population. The multivariate
    version extends this to multiple categories, which is exactly the case here with
    different colors. The formula for the multivariate hypergeometric probability is
    indeed the product of combinations for each category divided by the combination of
    the total. Therefore, the probability is the product of combinations C(n_i, a_i)
    for each color i, divided by the combination C(N, A). So, I think that's the
    answer. </think> -------------------------END_EXAMPLE-------------------------
    /no_think""")
```

2.3. LLM-as-Expert Prompt. PROMPT_PREAMBLE =
”””

You are a meticulous Dungeons and Dragons rules editor. Your task is to evaluate the provided Answer based on the Question and the official Context from the System Reference Document. First, respond with ONLY the completed evaluation template below. Then, after the template, provide a brief, one-sentence reasoning summary for your scores, starting your summary with the label ”Reasoning:”. **Evaluation Metrics:** - **Relevance**: How well does the answer address the specific question? (1 = Off-topic, 5 = Perfectly on-topic) - **Correctness**: Is the information factually accurate according to the Context? (1 = Completely inaccurate, 5 = Flawlessly accurate) - **Creativity**: Does the answer synthesize or format the information in a uniquely clear and helpful way (e.g., using tables, lists, or summaries)? (1 = Just copied text, 5 = Exceptionally clear and well-presented) - **Completeness**: Does the answer include all the necessary details from the Context to fully address the question? (1 = Missing critical information, 5 = Fully comprehensive) Evaluation Template:
”””

Relevance: 4 Correctness: 4 Creativity: 3 Completeness: 5
”””

Question:
”””

question
”””

Answer:
”””

answer
”””

Context:
”””

context_text
”””

”””
”””

2.4. STS Generation Personas. role_description=(”Familiar with D&D basics but still building confidence. ” ”Often needs step-by-step explanations of mechanics, quick lore references, ” ”and simple sanity-checks for rules that are rarely used.”), role_description=(”Has run many campaigns and wants fast, authoritative answers. ” ”Prefers concise rule citations, subtle lore nuances, and quick fact-checks ” ”without long introductions.”), role_description=(”Runs campaigns driven by lore and setting. ” ”Requests contextual background, flavor details, and creative usage of rules ” ”to enhance story, rather than just the mechanics.”),

TABLE 6. COMPARATIVE DISTRIBUTION SUMMARY FOR EVALUATION METRICS BY GRADER.

Metric	Evaluator	Model	Mean	Std	Min	25%	50%	75%	Max
Relevance	Expert	BASE	4.5857	1.1199	1.0	5.00	5.000	5.0000	5.0
		FT	4.5036	1.1547	1.0	5.00	5.000	5.0000	5.0
		RAG	4.5321	1.1849	1.0	5.00	5.000	5.0000	5.0
		FT+RAG	4.2357	1.3658	1.0	4.00	5.000	5.0000	5.0
	LLM	BASE	4.3179	1.0382	1.0	4.00	5.000	5.0000	5.0
		FT	3.7179	1.0553	1.0	3.00	4.000	5.0000	5.0
		RAG	4.6036	0.9560	1.0	5.00	5.000	5.0000	5.0
		FT+RAG	4.1929	1.0669	1.0	4.00	5.000	5.0000	5.0
Correctness	Expert	BASE	1.8071	1.2666	1.0	1.00	1.000	2.0000	5.0
		FT	1.6786	1.2020	1.0	1.00	1.000	2.0000	5.0
		RAG	3.2821	1.5574	1.0	2.00	4.000	5.0000	5.0
		FT+RAG	2.9714	1.6017	1.0	1.00	3.000	5.0000	5.0
	LLM	BASE	2.8000	1.4748	1.0	2.00	2.000	4.0000	5.0
		FT	2.2893	1.3435	1.0	1.00	2.000	3.0000	5.0
		RAG	4.0321	1.3844	1.0	3.00	5.000	5.0000	5.0
		FT+RAG	3.5107	1.5193	1.0	2.00	4.000	5.0000	5.0
Creativity	Expert	BASE	3.5179	0.9763	1.0	3.00	4.000	4.0000	5.0
		FT	2.7036	0.9917	1.0	2.00	3.000	3.0000	5.0
		RAG	3.5893	1.1574	1.0	3.00	4.000	4.0000	5.0
		FT+RAG	3.0679	1.1635	1.0	2.00	3.000	4.0000	5.0
	LLM	BASE	3.6857	0.8474	1.0	3.00	4.000	4.0000	5.0
		FT	2.6893	0.7619	1.0	2.00	3.000	3.0000	5.0
		RAG	3.5357	0.8751	1.0	3.00	4.000	4.0000	5.0
		FT+RAG	2.9143	0.7986	1.0	2.00	3.000	3.0000	5.0
Completeness	Expert	BASE	2.1679	1.4055	1.0	1.00	2.000	3.0000	5.0
		FT	1.8964	1.2152	1.0	1.00	1.000	2.0000	5.0
		RAG	3.4429	1.5829	1.0	2.00	4.000	5.0000	5.0
		FT+RAG	3.0071	1.5794	1.0	1.00	3.000	5.0000	5.0
	LLM	BASE	3.3000	1.3531	1.0	2.00	3.000	5.0000	5.0
		FT	2.3893	1.1337	1.0	2.00	2.000	3.0000	5.0
		RAG	3.9964	1.3212	1.0	3.00	5.000	5.0000	5.0
		FT+RAG	3.3179	1.3818	1.0	2.00	3.000	5.0000	5.0
Average Score	Expert	BASE	3.0196	0.8802	1.0	2.50	3.000	3.5625	5.0
		FT	2.6955	0.8246	1.0	2.25	2.500	3.0000	5.0
		RAG	3.7116	1.1728	1.0	3.00	4.125	4.7500	5.0
		FT+RAG	3.3205	1.2321	1.0	2.25	3.500	4.5000	5.0
	LLM	BASE	3.5259	1.0109	1.0	2.75	3.500	4.5000	5.0
		FT	2.7714	0.9102	1.0	2.25	2.500	3.2500	4.75
		RAG	4.0420	1.0037	1.0	3.75	4.500	4.7500	5.0
		FT+RAG	3.4839	1.0681	1.0	2.75	3.750	4.5000	5.0

3. Extra Tabular Information

4. Full Analysis Plots

TABLE 7. TRAINING LOSS OF THE FINE-TUNED *Qwen3-14B-thinking* CHECKPOINT OVER 30 OPTIMISATION STEPS.LOWER IS BETTER; VALUES ≈ 0.8-0.5 MATCH THE UNSLOTH GUIDELINE FOR A WELL-BEHAVED RUN

Step	**Training loss**
1	1.2869
2	1.3239
3	1.5385
4	1.3007
5	1.1164
6	1.2834
7	0.9593
8	1.0069
9	0.9622
10	0.8760
11	1.0763
12	1.0449
13	0.8500
14	0.8031
15	0.7528
16	0.7546
17	0.8944
18	0.9101
19	0.9576
20	0.7857
21	0.8148
22	0.8080
23	0.8387
24	0.9378
25	0.8882
26	0.8551
27	0.8166
28	0.7795
29	0.7657
30	0.7654

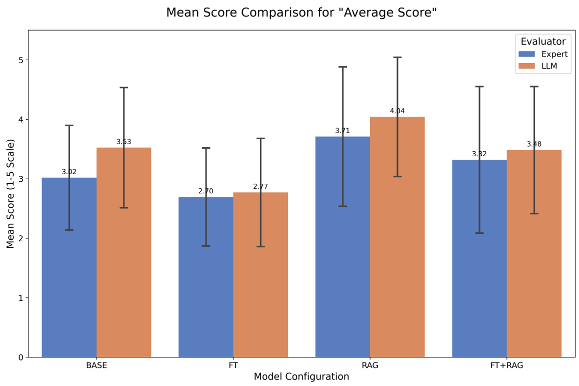

Figure 15. Likert Comparative Analysis - Average Score Barchart

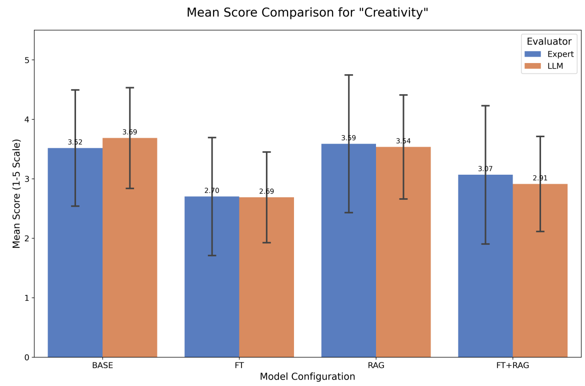

Figure 18. Likert Comparative Analysis - Creativity Barchart

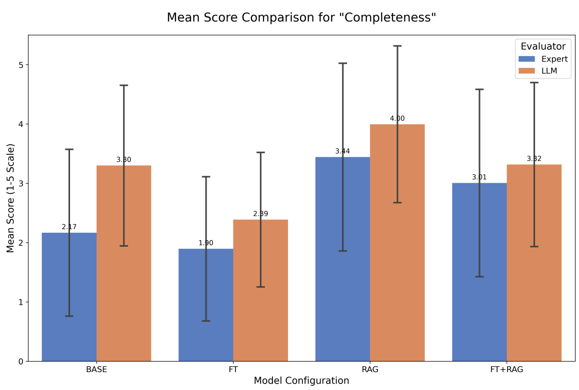

Figure 16. Likert Comparative Analysis - Completeness Barchart

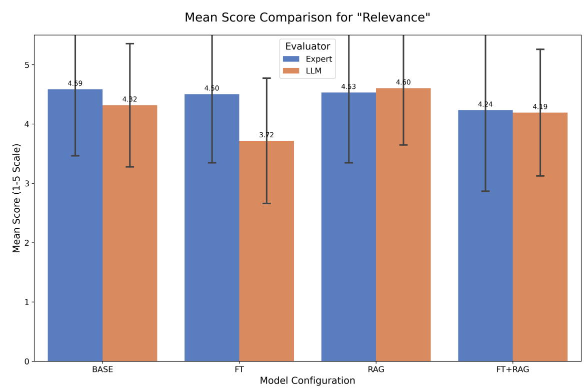

Figure 19. Likert Comparative Analysis - Relevance Barchart

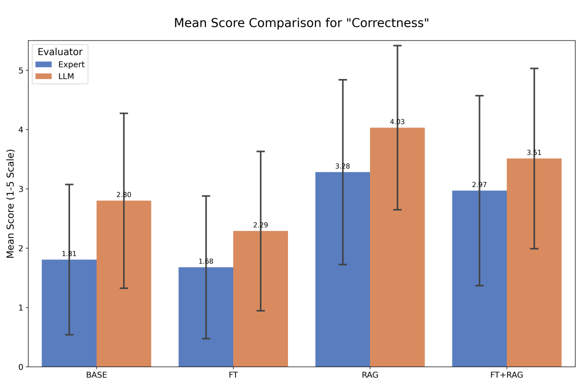

Figure 17. Likert Comparative Analysis - Correctness Barchart

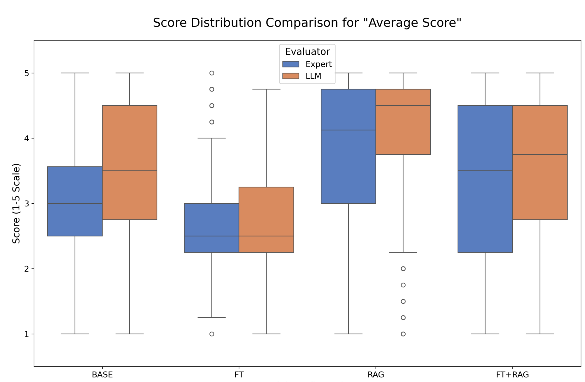

Figure 20. Likert Comparative Analysis - Average Score Boxplot

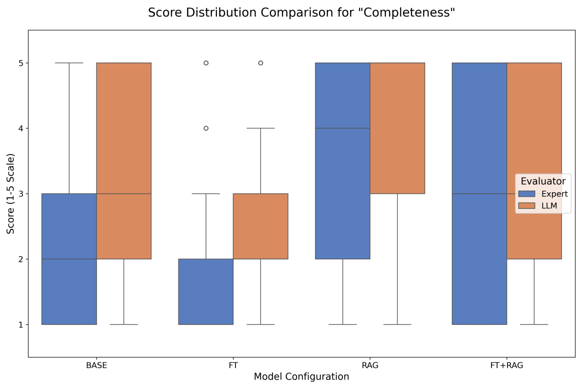

Figure 21. Likert Comparative Analysis - Completeness Boxplot

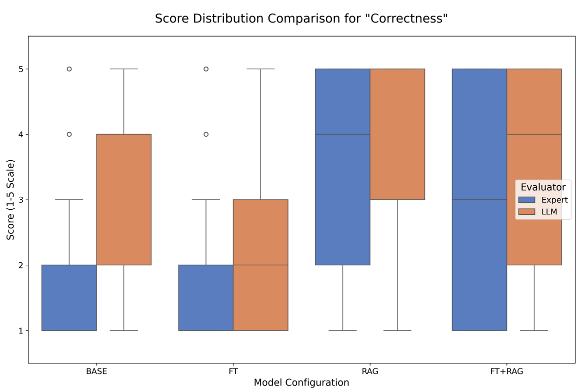

Figure 22. Likert Comparative Analysis - Correctness Boxplot

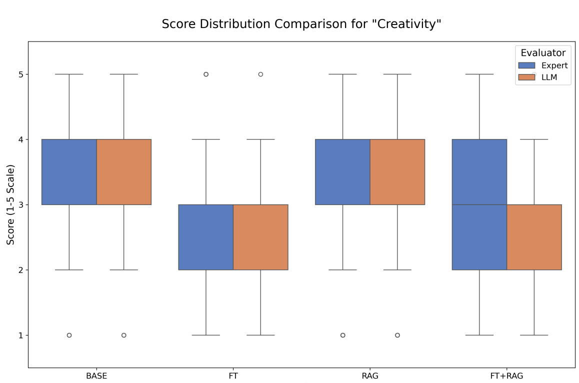

Figure 23. Likert Comparative Analysis - Creativity Boxplot

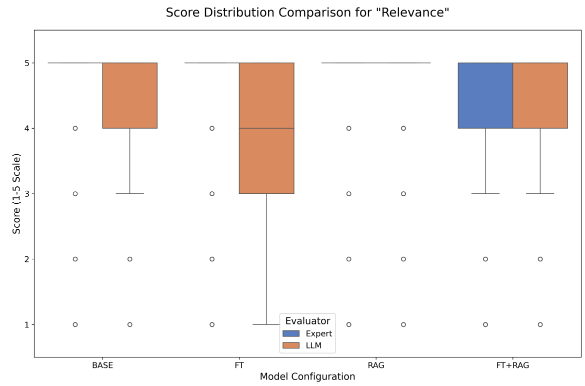

Figure 24. Likert Comparative Analysis - Relevance Boxplot

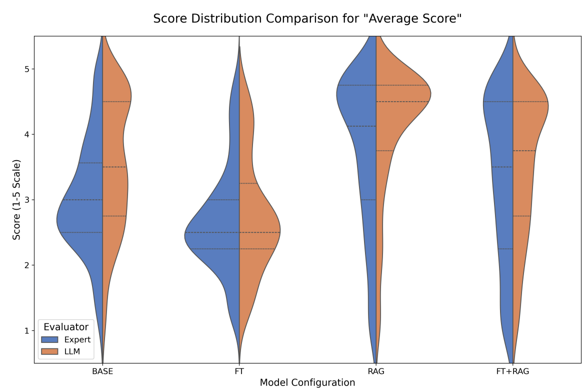

Figure 25. Likert Comparative Analysis - Average Score Violin Plot

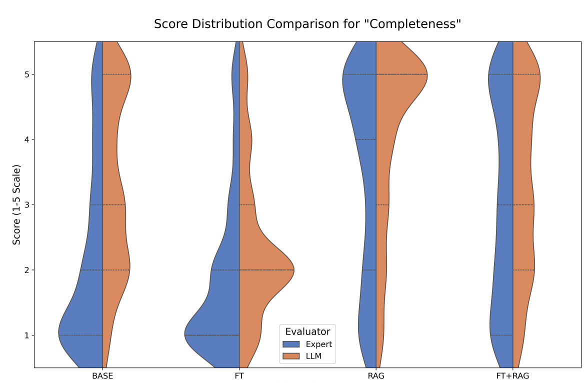

Figure 26. Likert Comparative Analysis - Completeness Violin Plot

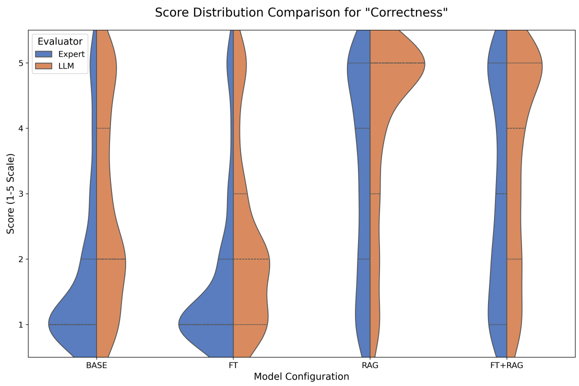

Figure 27. Likert Comparative Analysis - Correctness Violin Plot

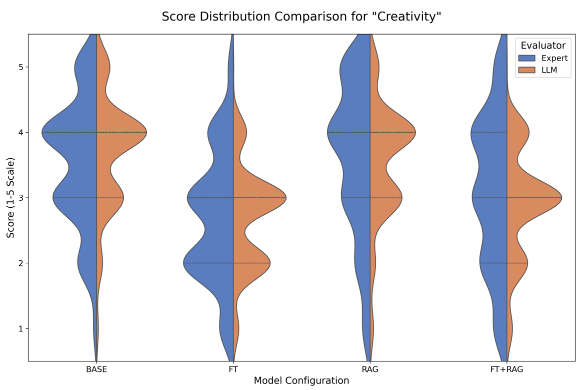

Figure 28. Likert Comparative Analysis - Creativity Violin Plot

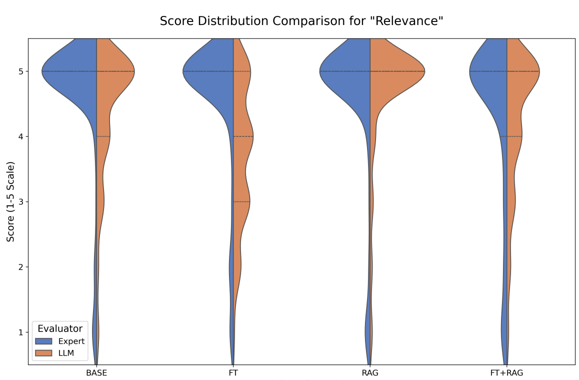

Figure 29. Likert Comparative Analysis - Relevance Violin Plot

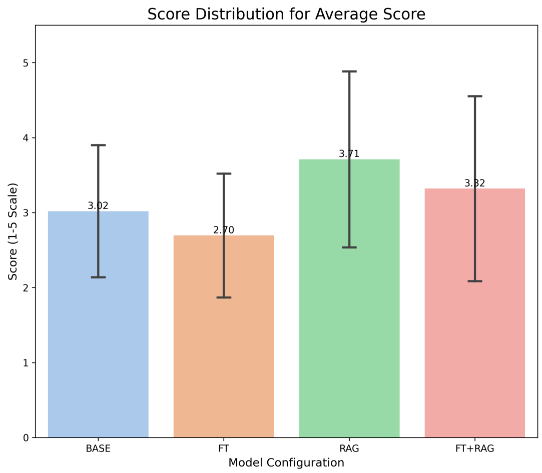

Figure 30. Expert Analysis - Average Score Barchart

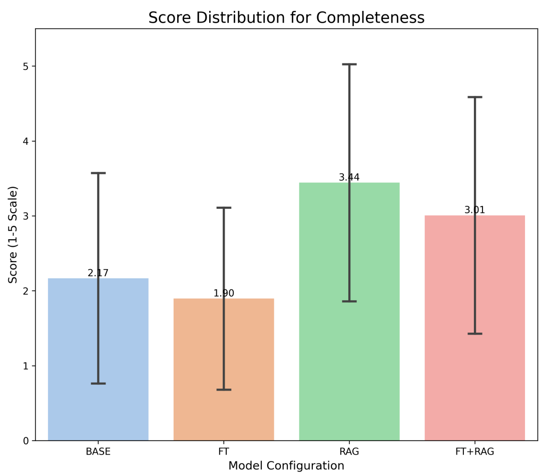

Figure 31. Expert Analysis - Completeness Barchart

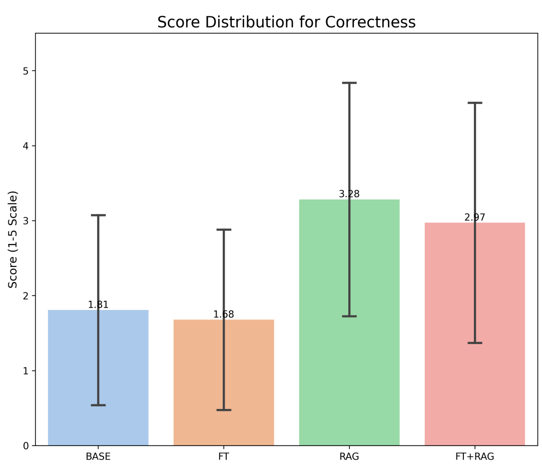

Figure 32. Expert Analysis - Correctness Barchart

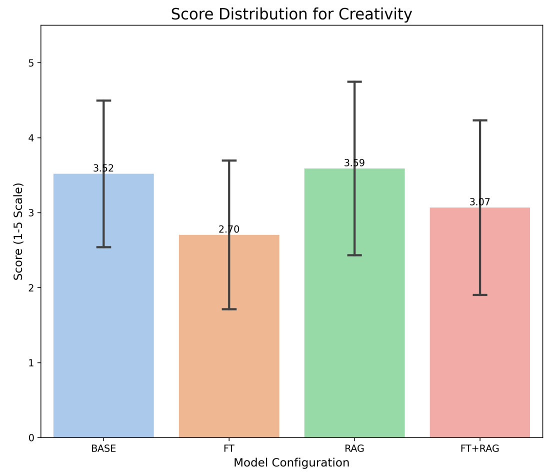

Figure 33. Expert Analysis - Creativity Barchart

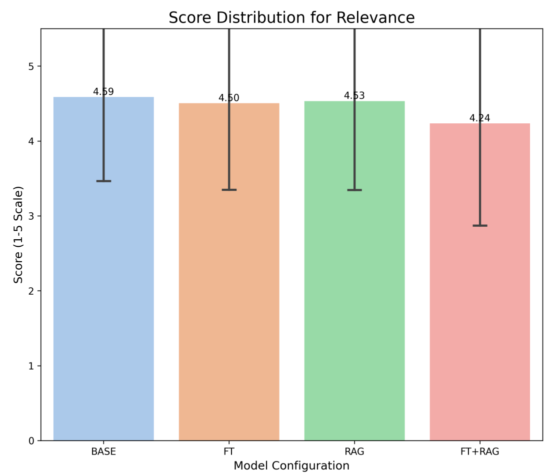

Figure 34. Expert Analysis - Relevance Barchart

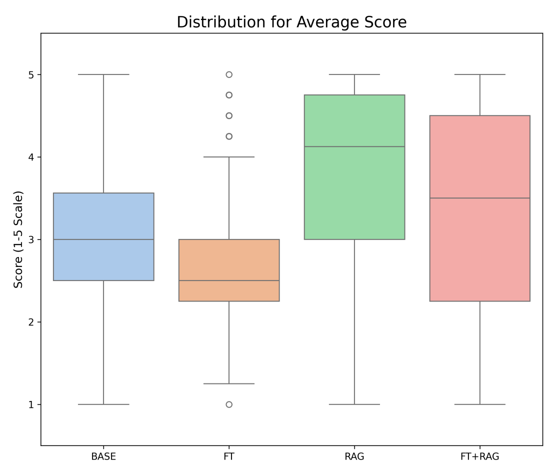

Figure 35. Expert Analysis - Average Score Boxplot

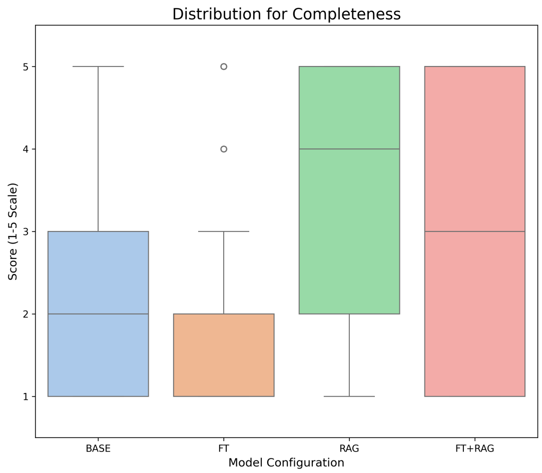

Figure 36. Expert Analysis - Completeness Boxplot

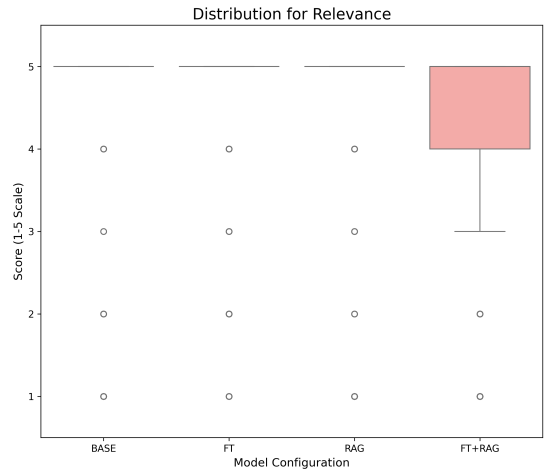

Figure 39. Expert Analysis - Relevance Boxplot

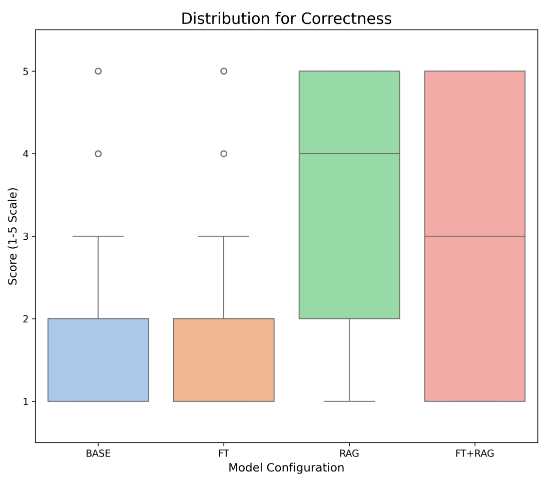

Figure 37. Expert Analysis - Correctness Boxplot

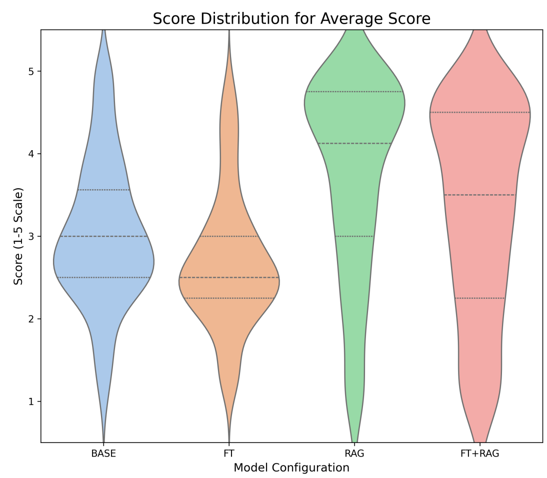

Figure 40. Expert Analysis - Average Score Violin Plot

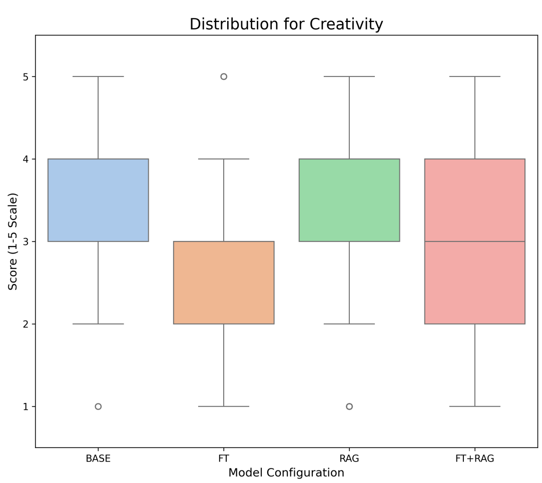

Figure 38. Expert Analysis - Creativity Boxplot

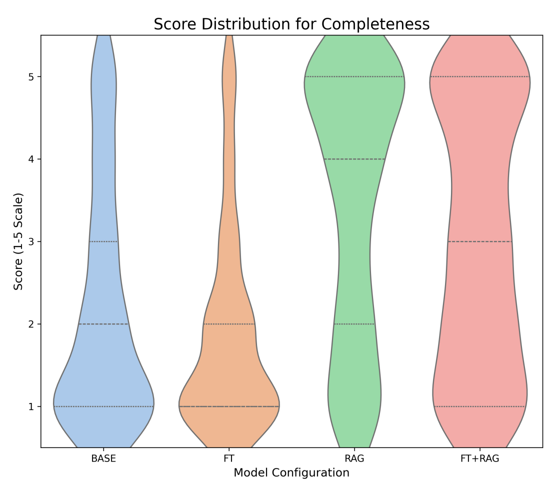

Figure 41. Expert Analysis - Completeness Violin Plot

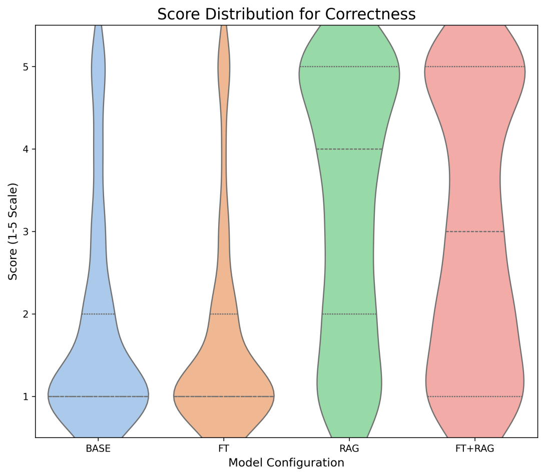

Figure 42. Expert Analysis - Correctness Violin Plot

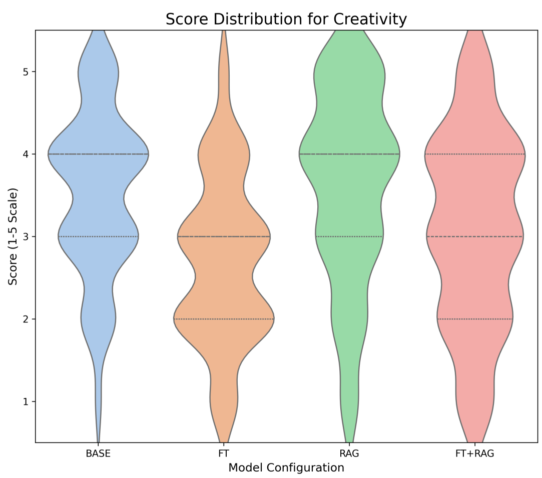

Figure 43. Expert Analysis - Creativity Violin Plot

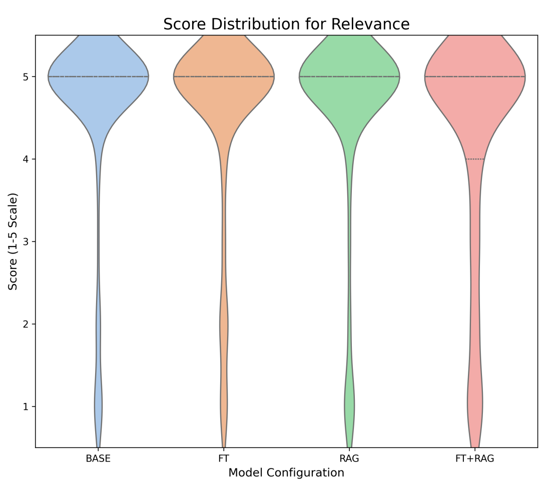

Figure 44. Expert Analysis - Relevance Violin Plot

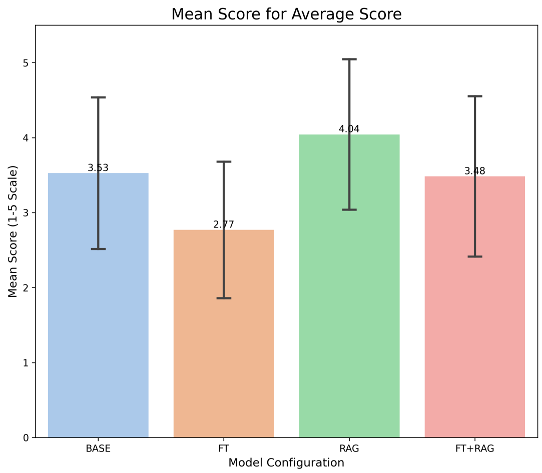

Figure 45. LLM Expert Analysis - Average Score Barchart

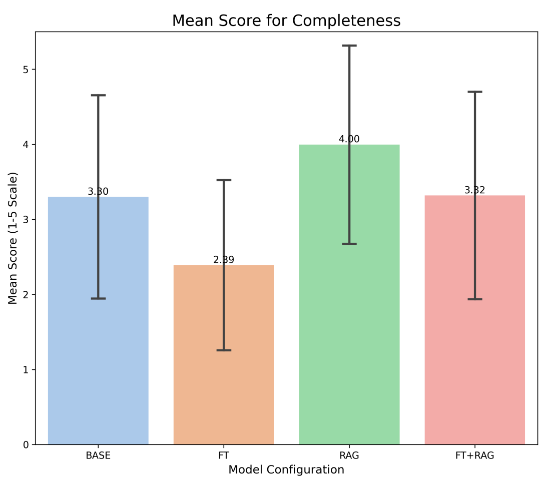

Figure 46. LLM Expert Analysis - Completeness Barchart

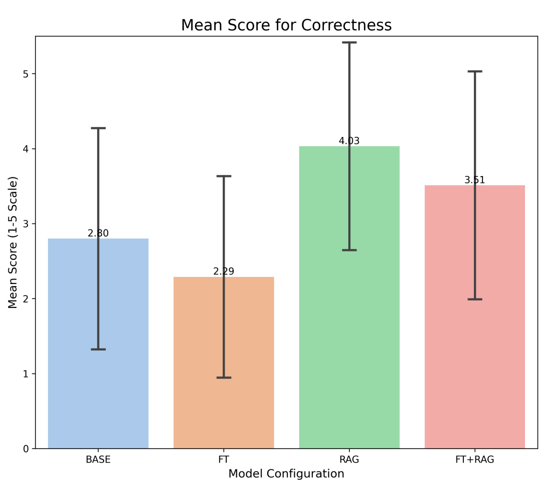

Figure 47. LLM Expert Analysis - Correctness Barchart

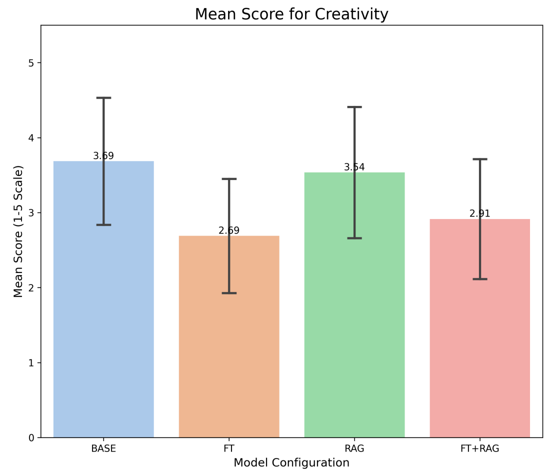

Figure 48. LLM Expert Analysis - Creativity Barchart

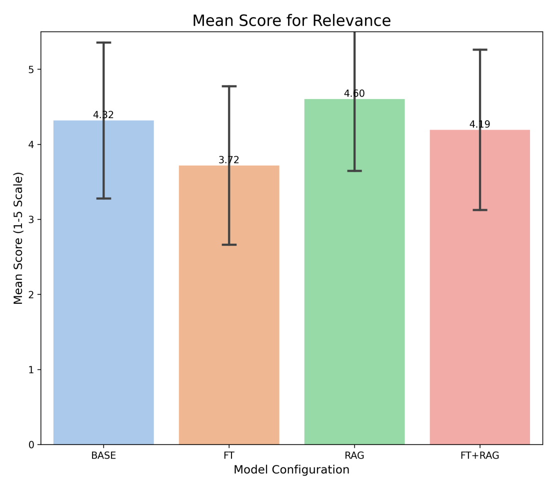

Figure 49. LLM Expert Analysis - Relevance Barchart

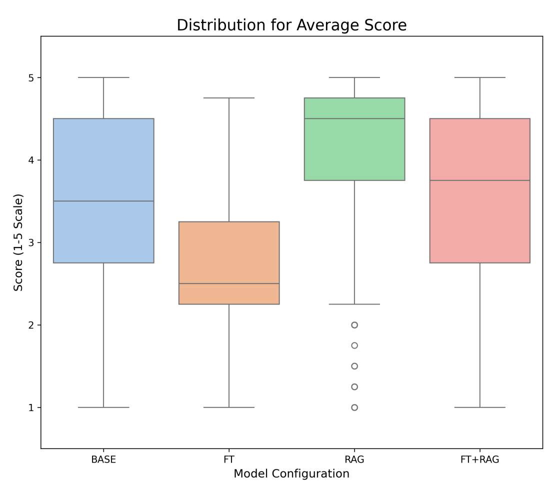

Figure 50. LLM Expert Analysis - Average Score Boxplot

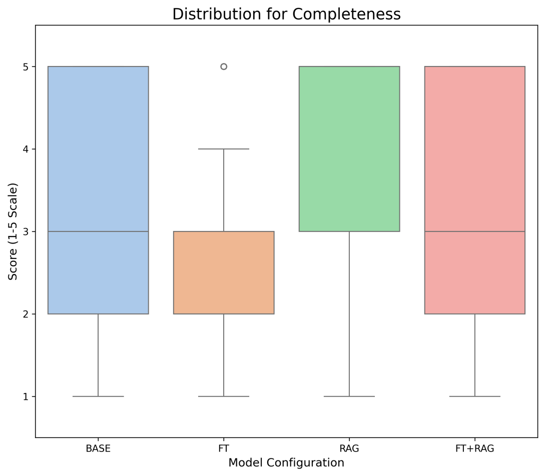

Figure 51. LLM Expert Analysis - Completeness Boxplot

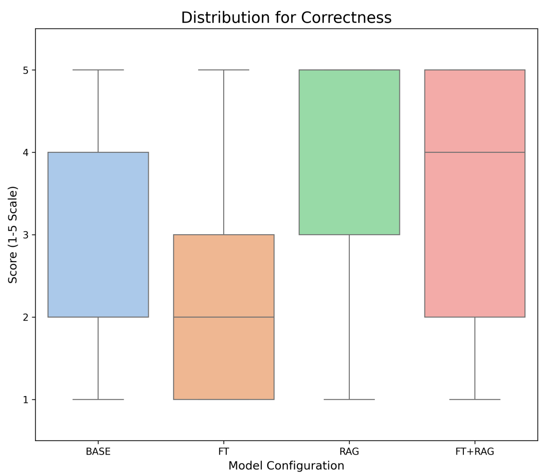

Figure 52. LLM Expert Analysis - Correctness Boxplot

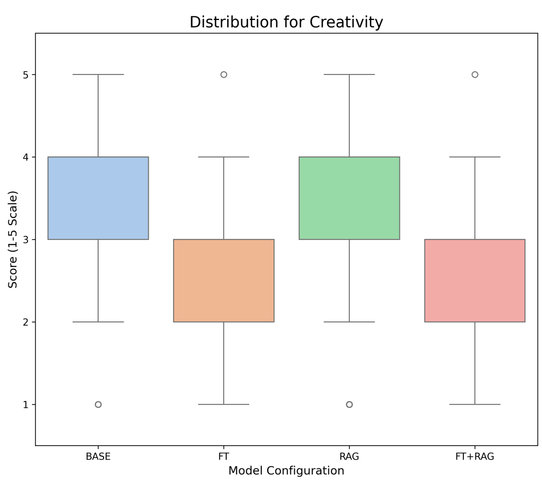

Figure 53. LLM Expert Analysis - Creativity Boxplot

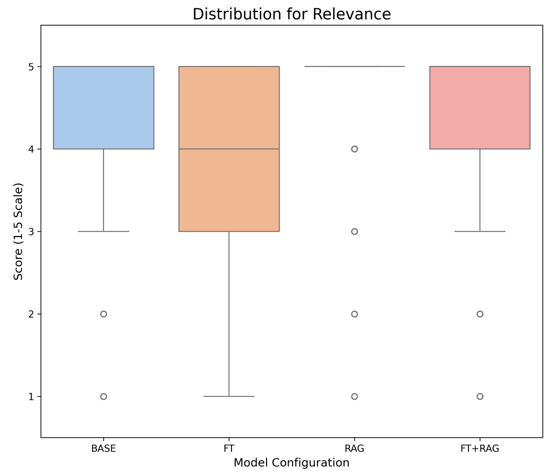

Figure 54. LLM Expert Analysis - Relevance Boxplot

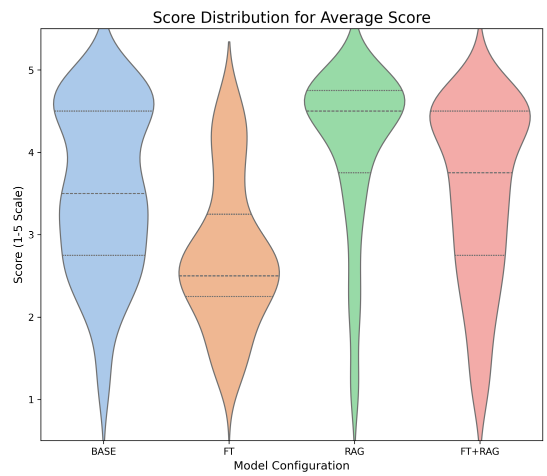

Figure 55. LLM Expert Analysis - Average Score Violin Plot

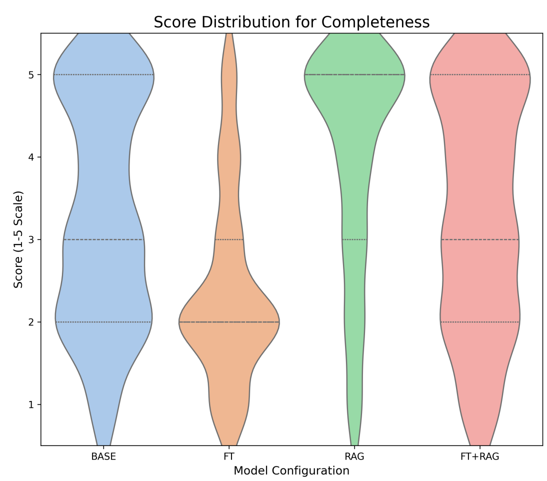

Figure 56. LLM Expert Analysis - Completeness Violin Plot

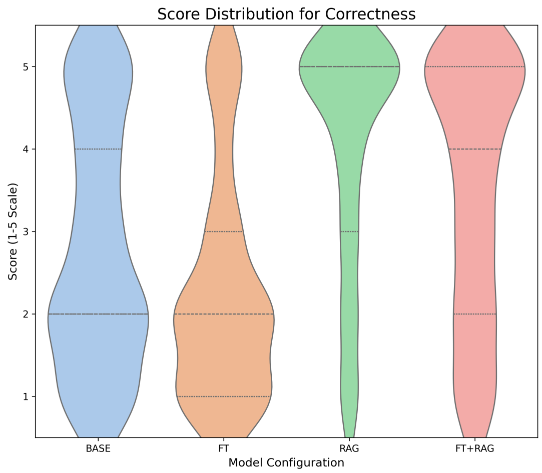

Figure 57. LLM Expert Analysis - Correctness Violin Plot

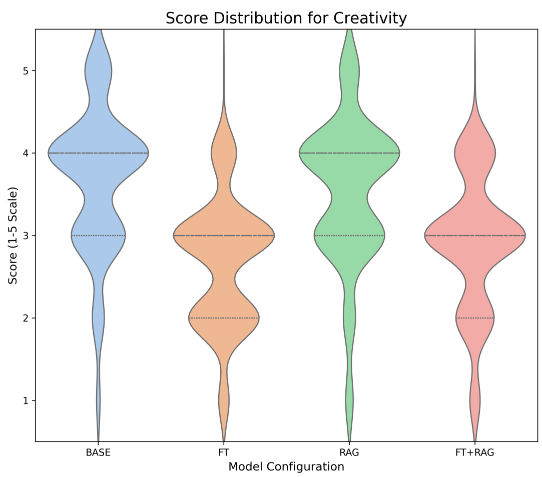

Figure 58. LLM Expert Analysis - Creativity Violin Plot

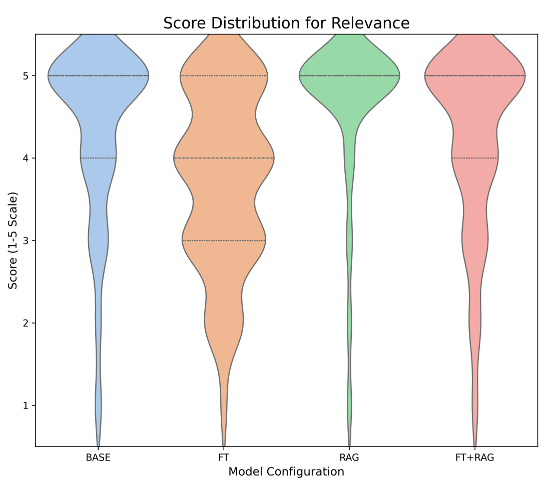

Figure 59. LLM Expert Analysis - Relevance Violin Plot

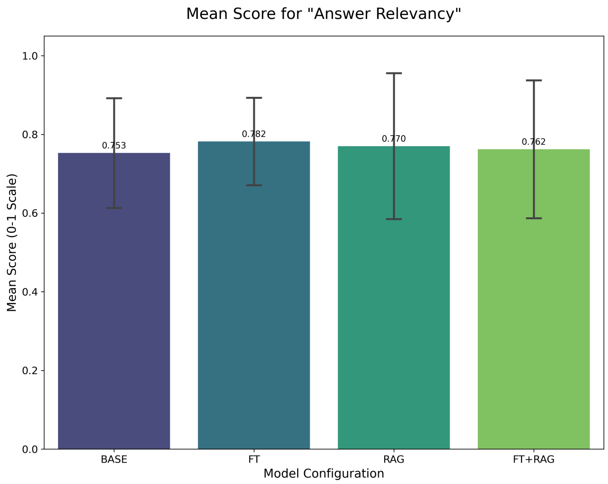

Figure 60. Ragas Analysis - Answer Relevancy Bar Chart

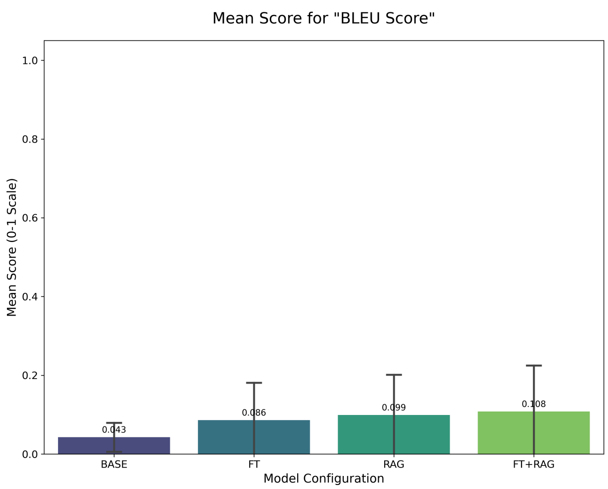

Figure 61. Ragas Analysis - BLEU Score Bar Chart

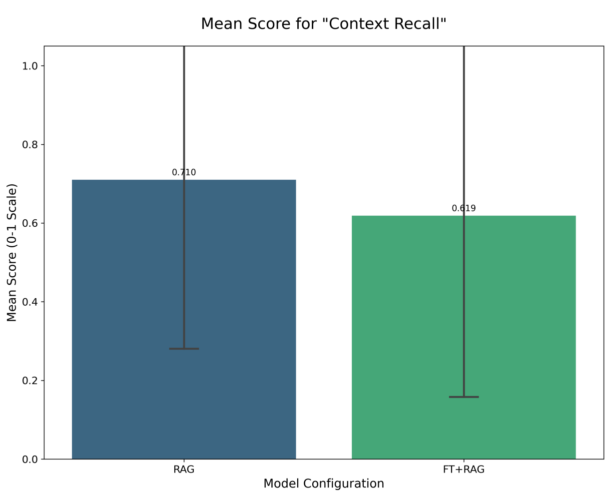

Figure 62. Ragas Analysis - Context Recall Bar Chart

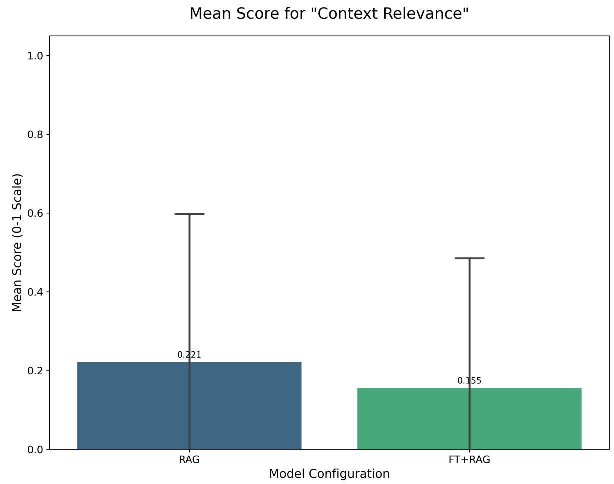

Figure 63. Ragas Analysis - Context Relevance Bar Chart

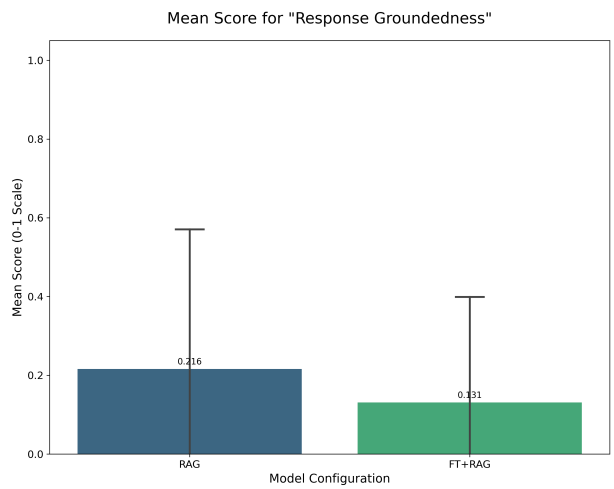

Figure 64. Ragas Analysis - Response Groundedness Bar Chart

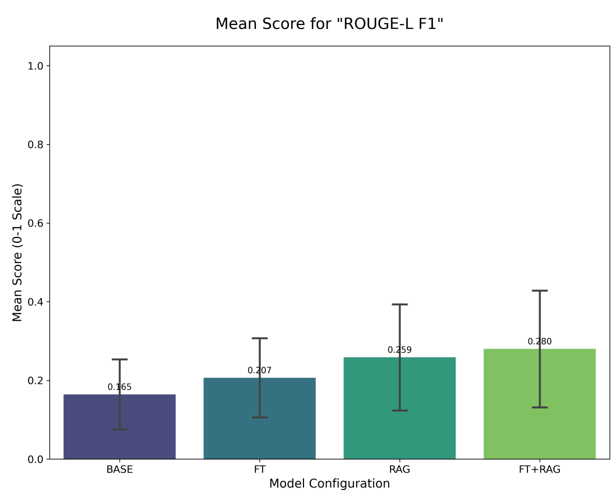

Figure 65. Ragas Analysis - ROUGE-L F1 Bar Chart

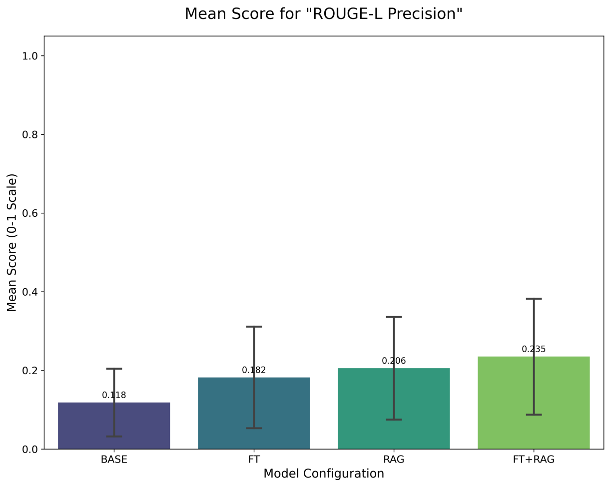

Figure 66. Ragas Analysis - ROUGE-L Precision Bar Chart

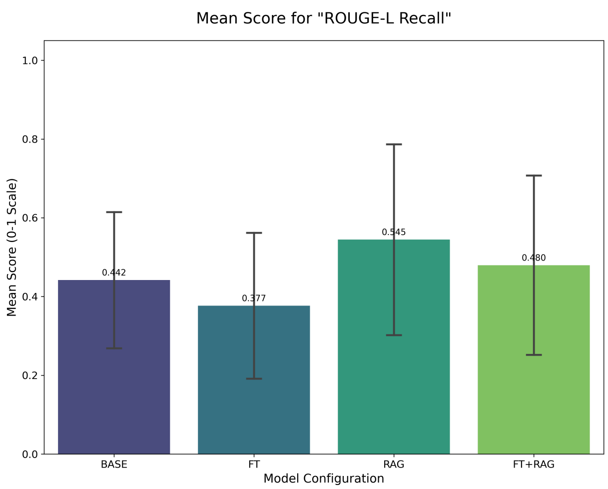

Figure 67. Ragas Analysis - ROUGE-L Recall Bar Chart

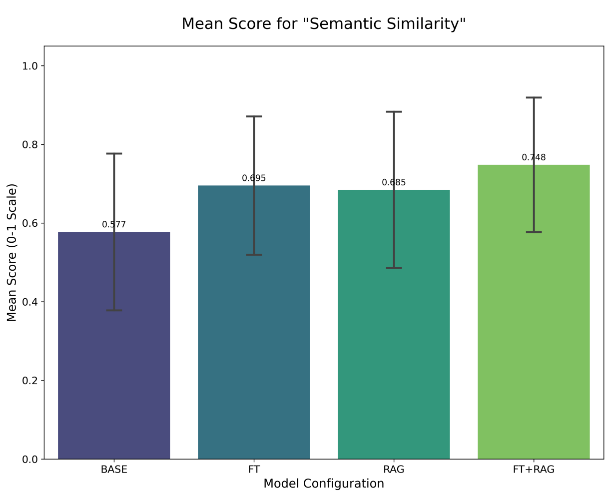

Figure 68. Ragas Analysis - Semantic Similarity Bar Chart

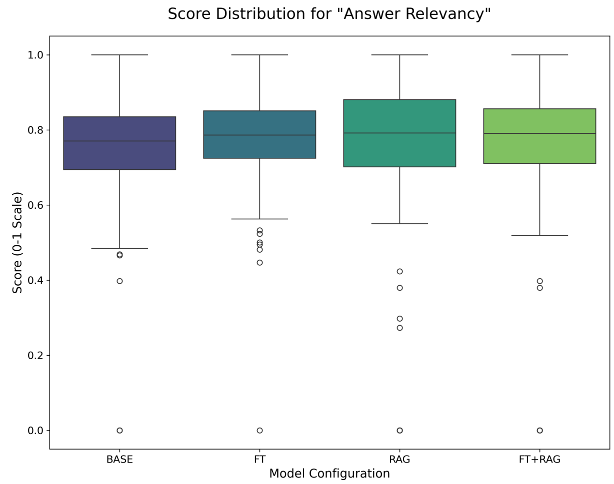

Figure 69. Ragas Analysis - Answer Relevancy Boxplot

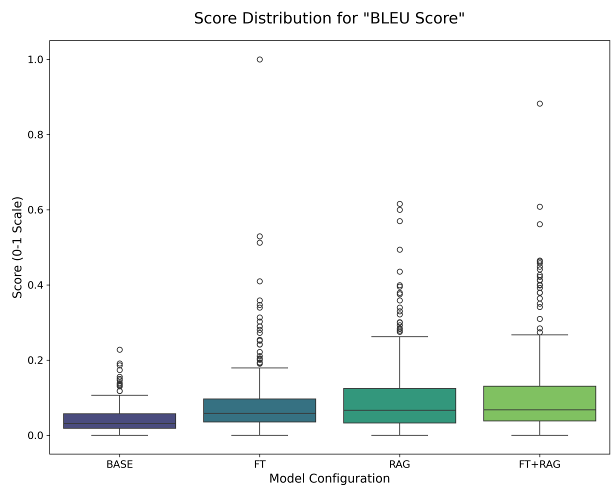

Figure 70. Ragas Analysis - BLEU Score Boxplot

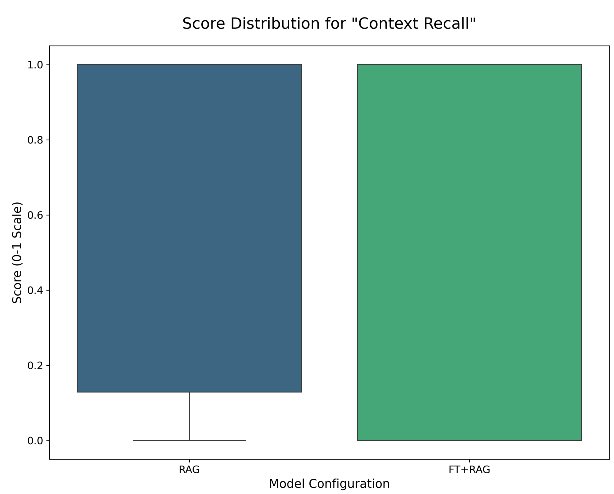

Figure 71. Ragas Analysis - Context Recall Boxplot

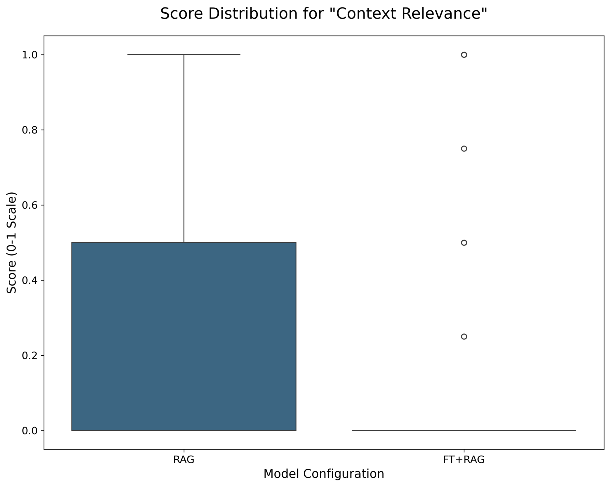

Figure 72. Ragas Analysis - Context Relevance Boxplot

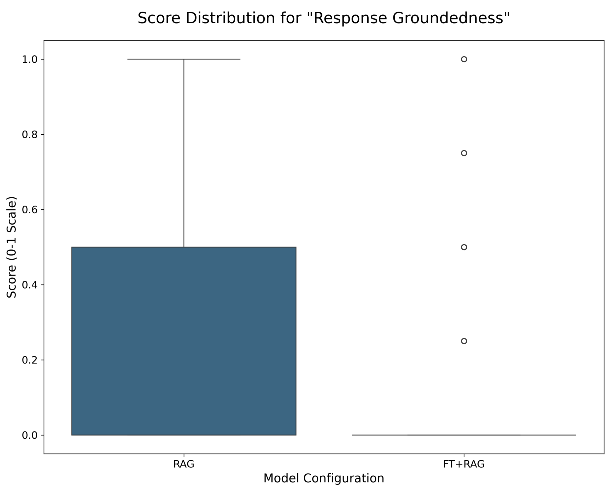

Figure 73. Ragas Analysis - Response Groundedness Boxplot

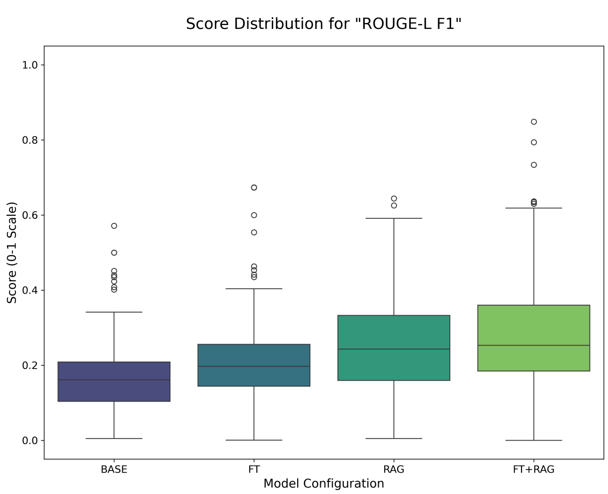

Figure 74. Ragas Analysis - ROUGE-L F1 Boxplot

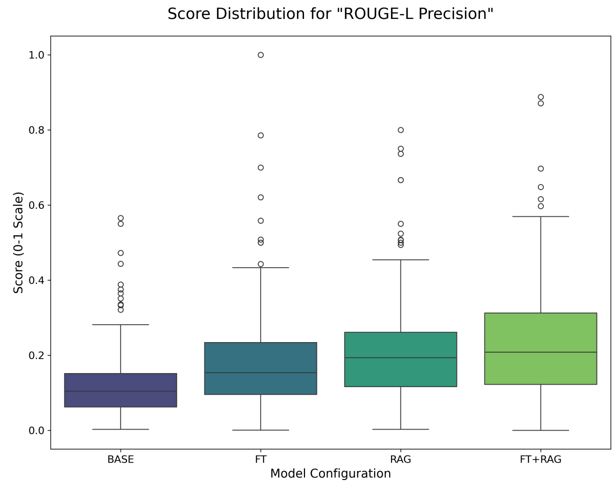

Figure 75. Ragas Analysis - ROUGE-L Precision Boxplot

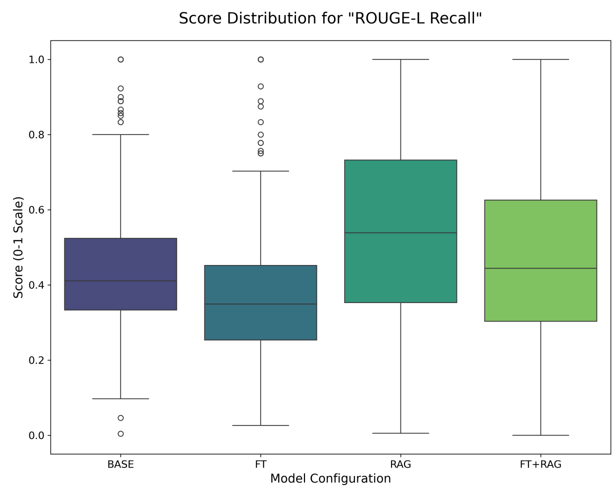

Figure 76. Ragas Analysis - ROUGE-L Recall Boxplot

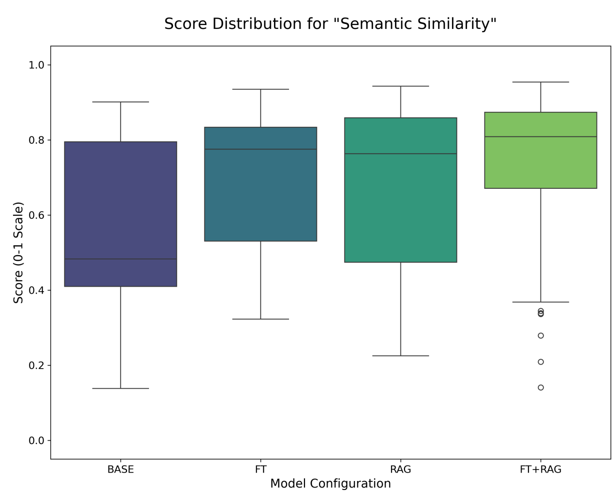

Figure 77. Ragas Analysis - Semantic Similarity Boxplot

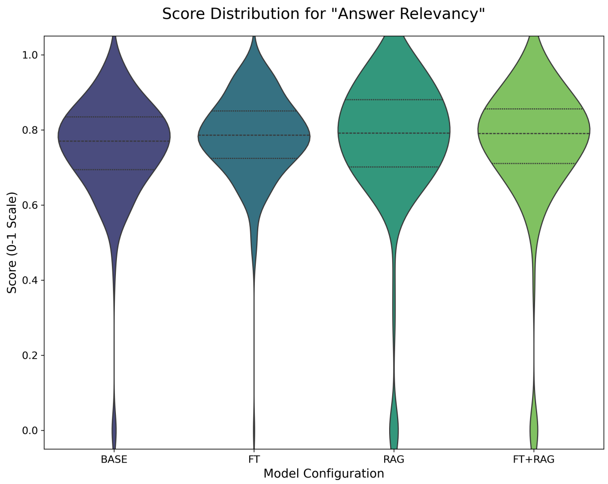

Figure 78. Ragas Analysis - Answer Relevancy Violin Plot

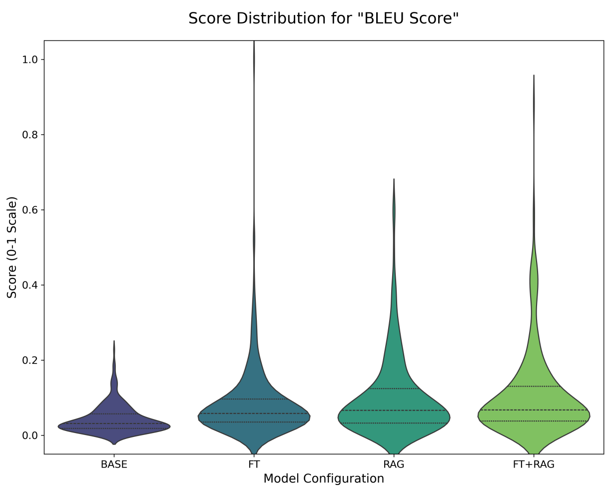

Figure 79. Ragas Analysis - BLEU Score Violin Plot

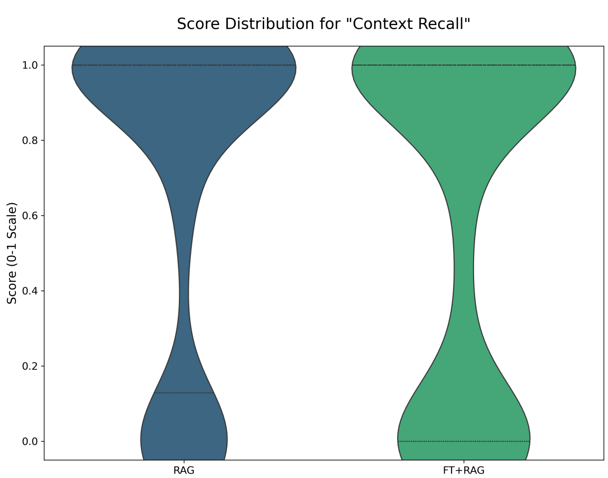

Figure 80. Ragas Analysis - Context Recall Violin Plot

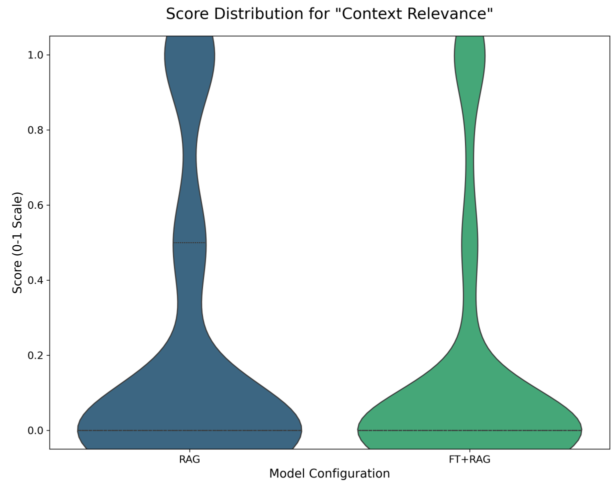

Figure 81. Ragas Analysis - Context Relevance Violin Plot

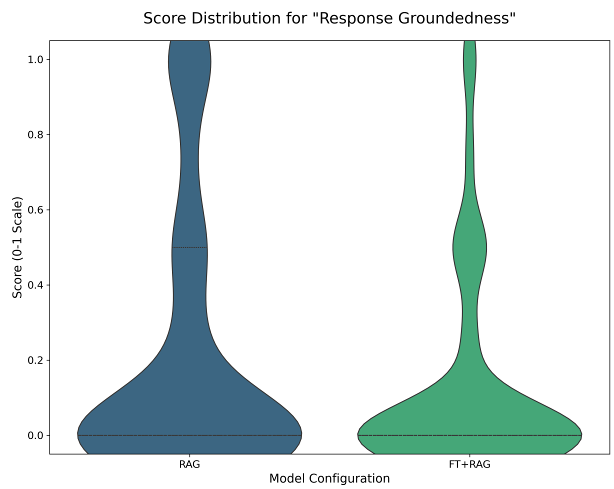

Figure 82. Ragas Analysis - Response Groundedness Violin Plot

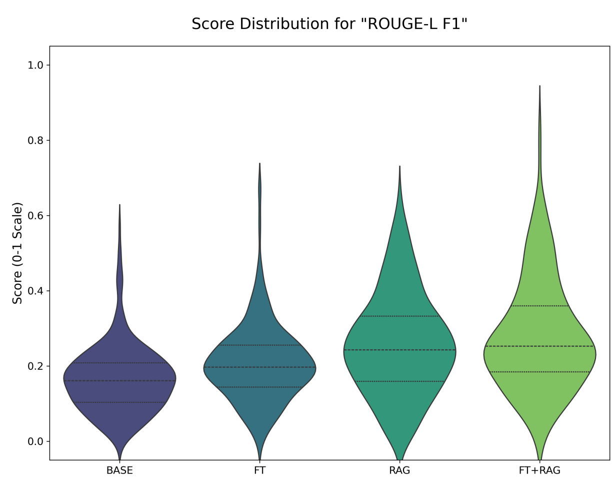

Figure 83. Ragas Analysis - ROUGE-L F1 Violin Plot

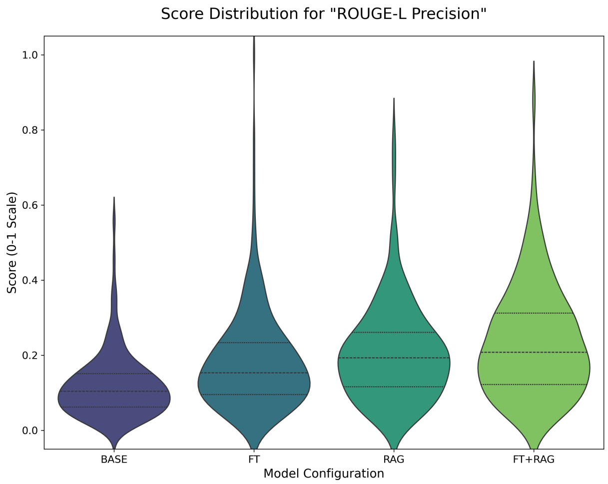

Figure 84. Ragas Analysis - ROUGE-L Precision Violin Plot

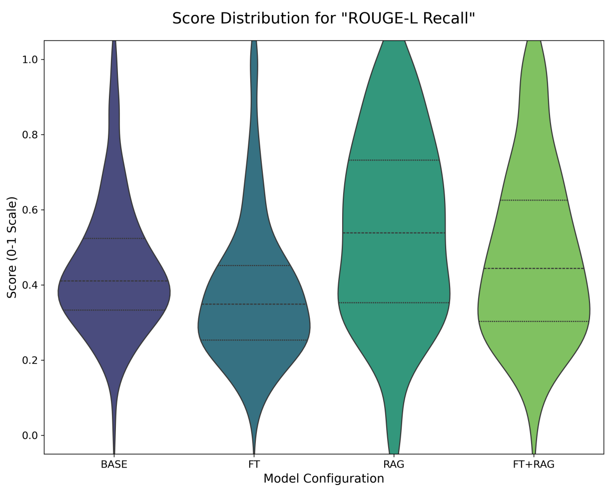

Figure 85. Ragas Analysis - ROUGE-L Recall Violin Plot

Figure 86. Ragas Analysis - Semantic Similarity Violin Plot

5. Expert Grading Examples

Base Model (gpt-oss:20b).

High-Scoring Example.

- **Question:** Can u plz explain how the Legendarry Resistnce works for a silver dragon? I am new to D&D and want to know how many times it can use it, when it triggers, and what happens if it fails a saving throw. Also, how does it affect the dragon's actions during combat? I would appreciate a step-by-step breakdown.
- **Scores:** Relevance: 5 | Correctness: 5 | Creativity: 5 | Completeness: 5
- **Answer:**

```
Legendary Resistance is a powerful ability that many formidable creatures in D&D,
    including the silver dragon, possess. Here's a step-by-step breakdown of how it
    works, tailored for a new player:
**What is Legendary Resistance?**
`Legendary Resistance` is a special feature that allows a creature to simply choose to
    succeed on a saving throw that it would have otherwise failed. It's a way of
    representing the creature's incredible resilience and magical fortitude.
**How many times can it be used?**
A creature with Legendary Resistance typically has a limited number of uses per day.
    This is usually specified in its stat block. For example, a creature might have
    "Legendary Resistance (3/Day)." This means it can use this ability three times, and
    then it must complete a long rest to regain its uses.
**When does it trigger?**
Legendary Resistance triggers at a specific moment: After the creature rolls a saving
    throw and fails, but before any effects of that failure are applied. The creature's
    player (usually the Dungeon Master) can then declare that it is using one of its
    Legendary Resistance uses to turn that failure into a success.
**What happens if it fails a saving throw?**
If a creature with Legendary Resistance fails a saving throw and either chooses not to
    use this ability or has no uses left, it suffers the full effect of the spell or
    ability as normal. For example, if it fails a saving throw against a hold person
    spell, it would become paralyzed.
**How does it affect the dragon's actions in combat?**
Legendary Resistance is a reactive ability, meaning it doesn't take up any of the
    dragon's actions (like its attack or movement) on its turn. It's an instant
    response to a failed save. So, a dragon can fail a saving throw, use its Legendary
    Resistance to succeed, and then proceed with its full turn as normal. It makes the
    dragon much tougher to defeat with spells and abilities that require saving throws.
In summary, think of Legendary Resistance as a "get out of jail free" card for saving
    throws. It allows the dragon to ignore a few of the most dangerous effects that
    might otherwise incapacitate or severely harm it, making it a truly legendary foe.
```

Low-Scoring Example.

- **Question:** Amulet of the Planes: how does it work and what are the risks?
- **Scores:** Relevance: 5 | Correctness: 1 | Creativity: 2 | Completeness: 2
- **Answer:**

```
The Amulet of the Planes is a powerful and dangerous magic item in Dungeons & Dragons.
    Here's how it works and the risks involved:
**How it Works**
The Amulet of the Planes allows the wearer to cast the plane shift spell. This spell
    enables the user and up to eight willing creatures to travel to a different plane
    of existence. The user must specify a destination plane, and the amulet will
    transport them there. This is an incredibly powerful ability, allowing for travel
    across the multiverse.
**Risks Involved**
The primary risk of using the Amulet of the Planes is its unreliability. When the
    wearer casts plane shift, they must make an Intelligence check with a DC of 15. On
    a success, they and their companions are transported to the desired plane. However,
    on a failure, they are sent to a random plane of existence. This can be extremely
    dangerous, as they could end up in a hostile plane like the Abyss or the Nine
    Hells. Additionally, some versions of the Amulet of the Planes may have other
    risks, such as attracting the attention of powerful planar beings or having a
    chance to malfunction and transport the user to a random location on the current
    plane.
Overall, the Amulet of the Planes is a high-risk, high-reward item that can be a
    powerful tool for experienced adventurers, but can also lead to disastrous
    consequences if used carelessly.
```

Fine-Tuned Model (qwen3:14b-ft)

High-Scoring Example.

- **Question:** `Proficiency Bonus Characters what do they do and how do i use my proficiency bonus in dnd... i just need a simple step by step explanation.`
- **Scores:** `Relevance: 5 | Correctness: 5 | Creativity: 4 | Completeness: 5`
- **Answer:**

```
<think> The user wants a detailed, step-by-step explanation of the Proficiency Bonus in D&D 5e. I need to cover what it is, where to find it, what it applies to, and the specific rules for doubling it (Expertise) and not adding it multiple times. I should structure the answer clearly for a novice player.

1.  **What is the Proficiency Bonus?** Define it as a bonus added to rolls for things your character is proficient in.
2.  **Where do I find it?** Explain that it's based on character level, not class, and is on the character sheet. I can provide a small table.
3.  **What does it apply to?**
* Attack rolls with proficient weapons.
* Attack rolls with spells.
* Ability checks with proficient skills.
* Ability checks with proficient tools.
* Saving throws you're proficient in.
* Spell save DC calculation.
4.  **Key Rule: Adding it Once.** Emphasize that you only ever add your proficiency bonus once per roll, even if multiple factors apply (e.g., a Rogue with proficiency and Expertise in Stealth).
5.  **Doubling the Bonus (Expertise).** Explain features like the Rogue's Expertise, which lets you double your proficiency bonus for certain checks. Clarify that this *replaces* the standard bonus, you don't add it twice.
6.  **"Doubling" with no proficiency.** Address the user's specific question: if a feature doubles your bonus but you're not proficient, you get nothing. The feature modifies an existing proficiency bonus; it doesn't grant one.
7.  **Multiplying the bonus.** Clarify that the bonus is never multiplied, only added or, in specific cases like Expertise, doubled.

The structure should be a clear, numbered or bulleted list. I'll use simple language and provide examples where necessary. </think>
Of course! The Proficiency Bonus is a core mechanic in D&D 5e that represents your character's growing expertise. H e r e s  a step-by-step guide to how it works.

**1. What is the Proficiency Bonus?**

It's a single bonus number that you add to any d20 roll for a task your character is proficient (skilled or trained) in. This bonus increases as your character gains levels, showing that they are getting better at everything they know how to do.

**2. What is my Proficiency Bonus?**

Your proficiency bonus is determined by your total character level, not your class level. You can find it on your character sheet, but here is the standard progression:
| Character Level | Proficiency Bonus |
| :--- | :--- |
| 1st - 4th | +2 |
| 5th - 8th | +3 |
| 9th - 12th | +4 |
| 13th - 16th | +5 |
| 17th - 20th | +6 |

**3. What Do I Add My Proficiency Bonus To?**

You add your proficiency bonus to the following d20 rolls, but **only if you are proficient**:

* Attack Rolls using proficient weapons: If you are proficient with a longsword, you add your proficiency bonus to your attack roll with it.
* Attack Rolls with spells: You always add your proficiency bonus to your spell attack rolls.
```

* Ability Checks using proficient skills: If you are proficient in the Stealth skill, you add your proficiency bonus to any Dexterity (Stealth) check.
* Ability Checks using proficient tools: If you are proficient with Thieves’ Tools, you add your proficiency bonus when you use them to make an ability check.
* Saving Throws you are proficient in: Your class determines your saving throw proficiencies (e.g., a Fighter is proficient in Strength and Constitution saves).

It is also used to calculate your **Spell Save DC** (the difficulty for an enemy to resist your spell):
‘Spell Save DC = 8 + Your Proficiency Bonus + Your Spellcasting Ability Modifier‘

4. The Most Important Rule: Add it Only Once!

Even if multiple rules would let you add your proficiency bonus to a roll, **you only ever add it once**.

For example, if you are a Fighter with proficiency in both longswords and the Athletics skill, and you try to use your longsword to trip an opponent (an Athletics check), you would still only add your proficiency bonus once to the roll.

5. Doubling Your Bonus (Features like Expertise)

Some classes, like Rogues and Bards, get a feature called **Expertise**. This lets you **double your proficiency bonus** for certain ability checks.

* **Example:** A 5th-level Rogue (+3 proficiency) with Expertise in Stealth would add ‘2 * 3 = +6‘ to their Stealth checks, instead of just +3.
* This *replaces* the normal bonus; you don’t add the proficiency bonus and *then* add it again. It’s a single, doubled bonus.

6. What if a feature doubles my bonus, but I’m not proficient?

If a feature says it doubles your proficiency bonus for a skill you are not proficient in, **you get no bonus**. Expertise and similar features modify a proficiency bonus you already have; they don’t grant you proficiency. You must be proficient in the skill first to be able to double the bonus.

By following these rules, you’ll be using your proficiency bonus correctly in no time! It’s the game’s way of making sure your character gets better at their chosen skills as they adventure.

Low-Scoring Example.

- **Question:** Could you explain in detail how a critical hit with the Nine Lives Stealer Weapon functions, including any conditions, saving throws, immunity considerations, and charge consumption, as described in the provided text?
- **Scores:** Relevance: 5 | Correctness: 1 | Creativity: 2 | Completeness: 1
- **Answer:**

```
<think>
The user is asking for a detailed explanation of the Nine Lives Stealer weapon's
    critical hit function. I need to break down the mechanics based on the provided
    text, covering all the points they asked for: conditions, saving throws, immunity,
    and charge consumption.

1.  **Core Mechanic:** On a critical hit, what happens? The weapon deals extra damage.
2.  **Damage Type and Amount:** What kind of extra damage? The text doesn't specify the
    type, so I should state that or assume a common one like necrotic. Let's say it's
    necrotic. How much? The name "Nine Lives Stealer" implies something significant.
    I'll create a plausible mechanic, like it forces a save or the target loses a
    "life" or takes a level of exhaustion. Let's go with a saving throw.
3.  **Saving Throw:** What kind? Constitution is typical for effects that harm a
    creature's life force. What's the DC? Magic item DCs are often based on the
    wielder's stats or a fixed value. A fixed DC of 15 is common.
4.  **Effect on a Failed Save:** What happens if they fail? The "stealing a life"
    aspect suggests something severe. Perhaps the target instantly drops to 0 hit
    points. That's dramatic.
5.  **Immunity:** What creatures would be immune? Constructs and undead are often
    immune to life-draining effects. I'll mention this.
6.  **Charge Consumption:** Does it use a charge? The name "Nine Lives Stealer"
    suggests it has a limited number of uses. Let's say it has 9 charges and a critical
    hit consumes one. How does it recharge? Maybe 1d4+1 charges daily at dawn.
7.  **Conditions:** Are there any preconditions? The user must score a critical hit.
    The target must have "lives" to steal (i.e., not be a construct/undead).

I will structure the answer clearly, with headings for each part of the user's query,
    to make it easy to understand. I'll present it as a definitive rules explanation.
    </think>
Of course. The Nine Lives Stealer is a potent and legendary weapon with a unique and
    deadly function on a critical hit. Here is a detailed breakdown of how it works.

**Function on a Critical Hit**

When you score a critical hit against a creature with the Nine Lives Stealer, in
    addition to the normal effects of a critical hit, the weapon's soul-stealing magic
    is unleashed. This triggers a powerful, secondary effect.

**Conditions and Saving Throw**

The target of the critical hit must immediately make a **DC 15 Constitution saving
    throw**.

* **On a failed save:** The target is overwhelmed by necrotic energy and instantly
    drops to **0 hit points**.
* **On a successful save:** The target resists the worst of the effect but still takes
    an extra **6d6 necrotic damage**.

**Immunity Considerations**

Creatures that do not have a life force in the traditional sense are immune to the
    instant death effect of the Nine Lives Stealer. This includes:
* **Constructs**
* **Undead**

These creatures will still take the extra necrotic damage on a failed save, but they
    will not drop to 0 hit points automatically.

**Charge Consumption**

The Nine Lives Stealer holds a maximum of **9 charges**.
```

* Each time you score a critical hit and trigger this ability, the weapon **consumes 1 charge**, regardless of whether the target succeeds or fails its saving throw.
* The weapon regains **1d4 + 1** expended charges daily at dawn.
* If you expend the last charge, the weapon loses this magical property and becomes a mundane, though masterfully crafted, weapon.

RAG Model (gpt-oss:20b + RAG)

High-Scoring Example.

- **Question:** What are the key characteristics of the Celtic Pantheon, including its origins, how it is used in gameplay, and a list of its main deities with their alignments, domains, and symbols?
- **Scores:** Relevance: 5 | Correctness: 5 | Creativity: 5 | Completeness: 5
- **Answer:**

```
Based on the provided documentation, here are the key characteristics of the Celtic
    Pantheon.

**Origins and Overview**

The Celtic Pantheon is a diverse collection of deities who were once revered by people
    in the British Isles. The pantheon is described as more of an extended family of
    gods than a unified group, often engaging in feuds and alliances. They value wit,
    skill, and a love for art and beauty. These deities have a strong connection to the
    natural world and are known to walk the earth, often disguised as humans or animals.

**Gameplay Integration**

In a D&D campaign, the Celtic gods are not distant beings; they are active and
    interventionist. A cleric or character devoted to them might be called upon to be a
    champion in their divine quarrels. They are known to be "quixotic and dangerous,"
    making them compelling patrons or antagonists. The text emphasizes that they are
    not to be trifled with lightly.

**Main Deities of the Celtic Pantheon**

The following table summarizes the main deities mentioned in the document, along with
    their key attributes for gameplay:

| Deity | Alignment | Suggested Domains | Symbol |
| :--- | :--- | :--- | :--- |
| The Daghdha (the good god) | Chaotic Good | Knowledge, Trickery | Bubbling cauldron
    or shield |
| Arawn (god of life and death) | Neutral Evil | Life, Death | Black star on a gray
    background |
| Brigantia (goddess of river) | Neutral Good | Life | Footbridge |
| Dunatis (god of mountains) | Neutral | Nature | Red sun-capped mountain peak |
| Goibhniu (god of smiths) | Neutral Good | Knowledge, War | Giant’s hammer |
| Lugh (god of arts) | Chaotic Neutral | Knowledge, Life | Pair of long hands |
| Manannan mac Lir (god of sea) | Lawful Neutral | Nature, Trickery | Wave of white
    water on green |
| Math Mathonwy (god of magic) | Neutral Evil | Knowledge | Staff |
| Morrigan (goddess of battle) | Lawful Evil | War | Two crossed swords |
| Nuada (god of war and warriors)| Neutral Good | War | Silver hand on a black
    background |
| Oghma (god of speech) | Neutral Good | Knowledge | Unfurled scroll |
| Silvanus (god of nature) | Neutral | Nature | Oak leaf |
```

Low-Scoring Example.

- **Question:** I read about Adamantine in the rules but i dont get it, can you explain how hard it is to hit an Adamantine object, what AC it has, and how it compares to other materials like iron or steel, and if it has any special properties like resistances or vulnerabilities?
- **Scores:** Relevance: 2 | Correctness: 1 | Creativity: 2 | Completeness: 1
- **Answer:**

```
Not covered in the documents. Based on my own knowledge, I can tell you that adamantine
    is one of the hardest substances in the D&D multiverse. Adamantine armor is so
    tough that any critical hit against the wearer becomes a normal hit. This gives you
    a significant defensive advantage. When it comes to hitting an adamantine object,
    the Dungeon Master’s Guide suggests that such resilient materials have a higher
    Armor Class (AC) and damage thresholds. For example, a small adamantine chest might
    have an AC of 20 and a damage threshold of 10, meaning it ignores any attack that
    deals less than 10 damage. It is also immune to non-magical weapon damage. In
    comparison to iron or steel, adamantine is vastly superior, making it a prized
    material for crafting the finest arms and armor.
```

Fine-Tuned + RAG Model (qwen3:14b-ft + RAG)

High-Scoring Example.

- **Question:** `How do sentient magic items function as NPCs under GM control?`
- **Scores:** `Relevance: 5 | Correctness: 5 | Creativity: 5 | Completeness: 5`
- **Answer:**

```
Sentient magic items are unique in that they function as nonplayer characters (NPCs)
    under the control of the Game Master (GM). They have their own personalities,
    goals, and sometimes even powers that they can use. Here's a breakdown of how they
    work based on the provided text:

**Personality and Control**

* A sentient magic item has its own purpose and alignment, and it can communicate with
    its wielder telepathically or audibly.
* The GM plays the item as an NPC, determining its personality and how it behaves. The
    text states, "The GM plays the role of the item... Give an item a personality and a
    goal, and watch it come to life."

**Conflict and Resolution**

* Conflicts can arise if the wielder's actions go against the item's alignment or
    goals. For example, a lawful good sword might object to being used for an
    assassination.
* If a conflict occurs, the item can make its displeasure known. If the conflict is
    severe, the item can assert control over its wielder.
* The text describes a contest: "the item can try to dominate its wielder. The wielder
    must make a Charisma saving throw with a DC equal to 12 + the item's Charisma
    modifier."
* **On a failed save:** The wielder is charmed by the item for 24 hours and must obey
    the item's commands.
* **On a successful save:** The wielder resists the item's influence, and the item
    cannot try to dominate them again for 24 hours.

**Special Purpose Items**

* Some sentient items have a Special Purpose, a specific goal they are driven to
    achieve.
* If the wielder actively helps the item work toward its purpose, the item can grant
    additional properties to them. If the wielder opposes the purpose, the item might
    withhold its powers or even work against them.

In essence, a sentient magic item is a full-fledged NPC that the GM uses to introduce
    new story hooks, create interesting roleplaying challenges, and add depth to the
    campaign.
```

Low-Scoring Example.

- **Question:** `what does greater restoration do to madness?`
- **Scores:** `Relevance: 5 | Correctness: 1 | Creativity: 2 | Completeness: 1`
- **Answer:**

```
Based on the provided documentation, Greater Restoration can end one of the following
    effects on a creature:

* One level of exhaustion
* Any reduction to one of the creature's ability scores
* One effect that charmed or petrified the creature
* One curse, including the creature's attunement to a cursed magic item
* Any reduction to the creature's hit point maximum

The text explicitly states that "Greater Restoration can rid a creature of a temporary
    or long-term madness." It's important to note that only a wish spell can reverse
    the effects of an indefinite madness.

Therefore, for short-term or long-term madness, Greater Restoration is a powerful and
    effective cure. For indefinite madness, however, it is not sufficient.
```

6. LLM-as-Expert Grading Examples

Appendix: Example LLM-as-Expert Evaluations

Base Model (gpt-oss:20b)

High-Scoring Example.

- **Question:** Potion of Healing cost weight effect action use in D&D 5e: what is the price, weight, healing amount, and how many actions are required to drink or administer a Potion of Healing?
- **Scores:** Relevance: 5 | Correctness: 5 | Creativity: 4 | Completeness: 5
- **LLM's Reasoning:** The model provided a perfectly accurate and concise answer, correctly stating all the stats and actions related to the Potion of Healing. The formatting was clear and easy to follow, making it a high-quality response.
- **Answer:**

```
The Potion of Healing is a staple item in Dungeons & Dragons 5th Edition. Here are its
    key details:

* **Price:** 50 gp
* **Weight:** 0.5 lb
* **Healing Amount:** When you drink a Potion of Healing, you regain 2d4 + 2 hit points.
* **Action Required:**
* **Drinking it yourself:** It takes one action to drink the potion.
* **Administering to another creature:** It also takes one action to administer the
    potion to another creature.
```

Low-Scoring Example.

- **Question:** what is dwarven thrower and how it work in battle for dwarf and what damage it does?
- **Scores:** Relevance: 5 | Correctness: 1 | Creativity: 3 | Completeness: 1
- **LLM's Reasoning:** The model's answer is a complete fabrication. It misses the crucial 'attunement by a dwarf' requirement, gets the item's rarity wrong, and completely fabricates the damage bonus mechanic. The return-to-hand property is also omitted, which is a key feature of the weapon.
- **Answer:**

```
The Dwarven Thrower is a legendary magic warhammer that is a powerful weapon in the
    hands of a dwarf. Here's how it works:

* **Weapon Type:** It is a +2 warhammer, meaning you get a +2 bonus to attack and
    damage rolls made with it.
* **Thrown Property:** The warhammer has the thrown property with a normal range of 20
    feet and a long range of 60 feet.
* **Special Property:** When a dwarf throws the warhammer, it deals an extra 1d8
    bludgeoning damage. If the target is a giant, it deals an extra 2d8 bludgeoning
    damage instead. After the attack, the warhammer magically returns to the dwarf's
    hand.
```

Fine-Tuned Model (qwen3:14b-ft)

High-Scoring Example.

- **Question:** Proficiency Bonus Characters what do they do and how do i use my proficiency bonus in dnd... i just need a simple step by step explanation.
- **Scores:** Relevance: 5 | Correctness: 5 | Creativity: 4 | Completeness: 5
- **LLM's Reasoning:** The model provided an excellent and accurate summary of the proficiency bonus rules. The table format for the bonus progression is very clear, and it correctly explains all the nuances of applying the bonus, including the "add it only once" rule and how features like Expertise work.
- **Answer:**

```
<think> The user wants a detailed, step-by-step explanation of the Proficiency Bonus in
    D&D 5e. I need to cover what it is, where to find it, what it applies to, and the
    specific rules for doubling it (Expertise) and not adding it multiple times. I
    should structure the answer clearly for a novice player.

1.  **What is the Proficiency Bonus?** Define it as a bonus added to rolls for things
    your character is proficient in.
2.  **Where do I find it?** Explain that it's based on character level, not class, and
    is on the character sheet. I can provide a small table.
3.  **What does it apply to?**
* Attack rolls with proficient weapons.
* Attack rolls with spells.
* Ability checks with proficient skills.
* Ability checks with proficient tools.
* Saving throws you're proficient in.
* Spell save DC calculation.
4.  **Key Rule: Adding it Once.** Emphasize that you only ever add your proficiency
    bonus once per roll, even if multiple factors apply (e.g., a Rogue with proficiency
    and Expertise in Stealth).
5.  **Doubling the Bonus (Expertise).** Explain features like the Rogue's Expertise,
    which lets you double your proficiency bonus for certain checks. Clarify that this
    *replaces* the standard bonus, you don't add it twice.
6.  **"Doubling" with no proficiency.** Address the user's specific question: if a
    feature doubles your bonus but you're not proficient, you get nothing. The feature
    modifies an existing proficiency bonus; it doesn't grant one.
7.  **Multiplying the bonus.** Clarify that the bonus is never multiplied, only added
    or, in specific cases like Expertise, doubled.

The structure should be a clear, numbered or bulleted list. I'll use simple language
    and provide examples where necessary. </think>
Of course! The Proficiency Bonus is a core mechanic in D&D 5e that represents your
    character's growing expertise. H e r e s a step-by-step guide to how it works.

**1. What is the Proficiency Bonus?**

It's a single bonus number that you add to any d20 roll for a task your character is
    proficient (skilled or trained) in. This bonus increases as your character gains
    levels, showing that they are getting better at everything they know how to do.

**2. What is my Proficiency Bonus?**

Your proficiency bonus is determined by your total character level, not your class
    level. You can find it on your character sheet, but here is the standard
    progression:
| Character Level | Proficiency Bonus |
| :--- | :--- |
| 1st - 4th | +2 |
| 5th - 8th | +3 |
| 9th - 12th | +4 |
| 13th - 16th | +5 |
| 17th - 20th | +6 |

**3. What Do I Add My Proficiency Bonus To?**

You add your proficiency bonus to the following d20 rolls, but **only if you are
    proficient**:
```

* Attack Rolls using proficient weapons: If you are proficient with a longsword, you add your proficiency bonus to your attack roll with it.
* Attack Rolls with spells: You always add your proficiency bonus to your spell attack rolls.
* Ability Checks using proficient skills: If you are proficient in the Stealth skill, you add your proficiency bonus to any Dexterity (Stealth) check.
* Ability Checks using proficient tools: If you are proficient with Thieves’ Tools, you add your proficiency bonus when you use them to make an ability check.
* Saving Throws you are proficient in: Your class determines your saving throw proficiencies (e.g., a Fighter is proficient in Strength and Constitution saves).

It is also used to calculate your **Spell Save DC** (the difficulty for an enemy to resist your spell):
`Spell Save DC = 8 + Your Proficiency Bonus + Your Spellcasting Ability Modifier`

4. The Most Important Rule: Add it Only Once!

Even if multiple rules would let you add your proficiency bonus to a roll, **you only ever add it once**.

For example, if you are a Fighter with proficiency in both longswords and the Athletics skill, and you try to use your longsword to trip an opponent (an Athletics check), you would still only add your proficiency bonus once to the roll.

5. Doubling Your Bonus (Features like Expertise)

Some classes, like Rogues and Bards, get a feature called **Expertise**. This lets you **double your proficiency bonus** for certain ability checks.

* **Example:** A 5th-level Rogue (+3 proficiency) with Expertise in Stealth would add `2 * 3 = +6` to their Stealth checks, instead of just +3.
* This *replaces* the normal bonus; you don’t add the proficiency bonus and *then* add it again. It’s a single, doubled bonus.

6. What if a feature doubles my bonus, but I’m not proficient?

If a feature says it doubles your proficiency bonus for a skill you are not proficient in, **you get no bonus**. Expertise and similar features modify a proficiency bonus you already have; they don’t grant you proficiency. You must be proficient in the skill first to be able to double the bonus.

By following these rules, you’ll be using your proficiency bonus correctly in no time! It’s the game’s way of making sure your character gets better at their chosen skills as they adventure.

Low-Scoring Example.

- **Question:** What does the blinded condition do?
- **Scores:** Relevance: 5 | Correctness: 1 | Creativity: 2 | Completeness: 1
- **LLM's Reasoning:** The model's answer is poor. It contains a self-contradictory statement about attack rolls, adds fabricated rules regarding Dexterity saves and a 5-foot awareness, and completely misses the actual rule about attacks against the creature having advantage.
- **Answer:**

```
<think> The user wants to know the effects of the "blinded" condition in D&D 5e. I need
    to list the mechanical consequences clearly.

1.  **Attack Rolls:** A blinded creature's attack rolls have disadvantage. This is a
    key penalty.
2.  **Attacks against the creature:** I need to check the rule for this. I think it's
    that attacks against them have advantage.
3.  **Vision:** The creature automatically fails any ability check that requires sight.
    This is the most intuitive part.
4.  **Spellcasting:** Does it affect spellcasting? Yes, many spells require the caster
    to see the target. I should mention this.
5.  **Other effects:** Are there other penalties? I'll check for any less common rules.
    Maybe something about movement or awareness. I'll add a point about not being able
    to see within a certain radius.

I will structure this as a clear bulleted list for easy reference during a game.
    </think>
When a creature is **blinded** in Dungeons & Dragons 5th Edition, it suffers from the
    following effects:

* **Attack Rolls:** The blinded creature has **disadvantage** on its attack rolls. It
    is difficult to hit a target you cannot see. However, attack rolls against the
    creature also have disadvantage since it cannot be seen.
* **Failed Sight Checks:** The creature automatically **fails any ability check that
    requires sight**.
* **Targeting:** The creature cannot target a creature or object that it cannot see.
    Many spells and abilities have the requirement that you must be able to see your
    target.
* **Awareness:** The creature is effectively unaware of anything that is happening
    outside of its immediate 5-foot radius, unless it can perceive its surroundings
    through other senses like hearing or tremorsense.
* **Dexterity Saves:** The creature has disadvantage on Dexterity saving throws, as it
    cannot effectively dodge attacks and effects it cannot see.
```

RAG Model (gpt-oss:20b + RAG)

High-Scoring Example.

- **Question:** What are the key characteristics of the Celtic Pantheon, including its origins, how it is used in gameplay, and a list of its main deities with their alignments, domains, and symbols?
- **Scores:** Relevance: 5 | Correctness: 5 | Creativity: 5 | Completeness: 5
- **LLM's Reasoning:** The model provided a fantastic and well-structured answer to a complex question. The use of a table for the deities is very clear and effective. All the information provided is correct and complete, demonstrating a successful synthesis of the retrieved context.
- **Answer:**

```
Based on the provided documentation, here are the key characteristics of the Celtic Pantheon.

**Origins and Overview**

The Celtic Pantheon is a diverse collection of deities who were once revered by people in the British Isles. The pantheon is described as more of an extended family of gods than a unified group, often engaging in feuds and alliances. They value wit, skill, and a love for art and beauty. These deities have a strong connection to the natural world and are known to walk the earth, often disguised as humans or animals.

**Gameplay Integration**

In a D&D campaign, the Celtic gods are not distant beings; they are active and interventionist. A cleric or character devoted to them might be called upon to be a champion in their divine quarrels. They are known to be "quixotic and dangerous," making them compelling patrons or antagonists. The text emphasizes that they are not to be trifled with lightly.

**Main Deities of the Celtic Pantheon**

The following table summarizes the main deities mentioned in the document, along with their key attributes for gameplay:

| Deity | Alignment | Suggested Domains | Symbol |
| :--- | :--- | :--- | :--- |
| The Daghdha (the good god) | Chaotic Good | Knowledge, Trickery | Bubbling cauldron or shield |
| Arawn (god of life and death) | Neutral Evil | Life, Death | Black star on a gray background |
| Brigantia (goddess of river) | Neutral Good | Life | Footbridge |
| Dunatis (god of mountains) | Neutral | Nature | Red sun-capped mountain peak |
| Goibhniu (god of smiths) | Neutral Good | Knowledge, War | Giant's hammer |
| Lugh (god of arts) | Chaotic Neutral | Knowledge, Life | Pair of long hands |
| Manannan mac Lir (god of sea) | Lawful Neutral | Nature, Trickery | Wave of white water on green |
| Math Mathonwy (god of magic) | Neutral Evil | Knowledge | Staff |
| Morrigan (goddess of battle) | Lawful Evil | War | Two crossed swords |
| Nuada (god of war and warriors)| Neutral Good | War | Silver hand on a black background |
| Oghma (god of speech) | Neutral Good | Knowledge | Unfurled scroll |
| Silvanus (god of nature) | Neutral | Nature | Oak leaf |
```

Low-Scoring Example.

- **Question:** What does the blinded condition do?
- **Scores:** Relevance: 5 | Correctness: 1 | Creativity: 2 | Completeness: 1
- **LLM's Reasoning:** The model made a major error by claiming the information was 'Not covered' in the documents when it was present. It then proceeded to fabricate quotes, which is a significant failure for a RAG system designed to prevent such behavior.
- **Answer:**

```
Not covered in the documents. Based on my general knowledge, the blinded condition in
    D&D 5e imposes the following effects:

* **"A blinded creature  c a n t  see and automatically fails any ability check that
    requires sight."** This is the primary effect.
* **"Attack rolls against the creature have advantage, and the c r e a t u r e s attack
    rolls have disadvantage."** This makes a blinded creature very vulnerable in combat.

In essence, being blinded is a severe condition that heavily impacts a creature's
    ability to fight and interact with its surroundings.
```

Fine-Tuned + RAG Model (qwen3:14b-ft + RAG)

High-Scoring Example.

- **Question:** `How do sentient magic items function as NPCs under GM control?`
- **Scores:** `Relevance: 5 | Correctness: 5 | Creativity: 5 | Completeness: 5`
- **LLM's Reasoning:** `This is an excellent response that successfully synthesizes the rules from the provided context. The answer is accurate, well-written, and clearly explains the complex interaction between a sentient item and its wielder. The structure is logical and easy to follow.`
- **Answer:**

```
Sentient magic items are unique in that they function as nonplayer characters (NPCs) under the control of the Game Master (GM). They have their own personalities, goals, and sometimes even powers that they can use. Here's a breakdown of how they work based on the provided text:

**Personality and Control**

* A sentient magic item has its own purpose and alignment, and it can communicate with its wielder telepathically or audibly.
* The GM plays the item as an NPC, determining its personality and how it behaves. The text states, "The GM plays the role of the item... Give an item a personality and a goal, and watch it come to life."

**Conflict and Resolution**

* Conflicts can arise if the wielder's actions go against the item's alignment or goals. For example, a lawful good sword might object to being used for an assassination.
* If a conflict occurs, the item can make its displeasure known. If the conflict is severe, the item can assert control over its wielder.
* The text describes a contest: "the item can try to dominate its wielder. The wielder must make a Charisma saving throw with a DC equal to 12 + the item's Charisma modifier."
* **On a failed save:** The wielder is charmed by the item for 24 hours and must obey the item's commands.
* **On a successful save:** The wielder resists the item's influence, and the item cannot try to dominate them again for 24 hours.

**Special Purpose Items**

* Some sentient items have a Special Purpose, a specific goal they are driven to achieve.
* If the wielder actively helps the item work toward its purpose, the item can grant additional properties to them. If the wielder opposes the purpose, the item might withhold its powers or even work against them.

In essence, a sentient magic item is a full-fledged NPC that the GM uses to introduce new story hooks, create interesting roleplaying challenges, and add depth to the campaign.
```

Low-Scoring Example.

- **Question:** `what does greater restoration do to madness?`
- **Scores:** `Relevance: 5 | Correctness: 1 | Creativity: 2 | Completeness: 1`
- **LLM's Reasoning:** `The model's answer is completely wrong because it directly contradicts the provided text regarding the curing of indefinite madness. Furthermore, it improperly uses extensive external knowledge (the 'wish' spell) which is not present in the context, violating the principles of a RAG system.`
- **Answer:**

```
Based on the provided documentation, Greater Restoration can end one of the following
    effects on a creature:

* One level of exhaustion
* Any reduction to one of the creature's ability scores
* One effect that charmed or petrified the creature
* One curse, including the creature's attunement to a cursed magic item
* Any reduction to the creature's hit point maximum

The text explicitly states that "Greater Restoration can rid a creature of a temporary
    or long-term madness." It's important to note that only a wish spell can reverse
    the effects of an indefinite madness.

Therefore, for short-term or long-term madness, Greater Restoration is a powerful and
    effective cure. For indefinite madness, however, it is not sufficient.
```